HIKET — Bayesian Calibration of Forest SOC Models

Methods and pipeline documentation

Lorenzo Menichetti

July 2026

Contents

1 Overview

1.1 Research question

Finnish forest soils showed a positive trend in mineral SOC stocks across the first two systematic measurement campaigns — VMI8 (1985–86) and Biosoil (2006). This accumulation is what the forest-sink assumptions were built on, and its attribution is politically consequential: Finland reports the soil as a genuine sink, whereas comparable northern countries do not. The most recent campaign (Komeetta, 2024) — still under review during HIKET — appears to corroborate a *saturation* of this sink: the increase seen between the first two campaigns does not continue at the same rate.

Yasso — the model family underpinning Finland's soil carbon reporting — tends, when run with published parameter sets or calibrated under standard steady-state assumptions, to predict a saturation of the soil sink that is difficult to reconcile with the accumulation observed between the first two campaigns. (Other models would plausibly behave similarly, though this has not been tested here.) Because Finland's climate policy is built on the forest sink, the prospect that this sink is saturating is consequential — and motivates testing whether the predicted saturation is genuine or a calibration artefact. This raises the central question of HIKET:

> *Do divergent model projections of Finnish forest SOC — including apparent saturation — reflect genuine structural differences between models, or artefacts of how the models are calibrated?*

HIKET addresses this by calibrating six models against the same Finnish data under identical likelihood, error, and filtering assumptions, the same litter-input estimates and climate forcing, and — crucially — the same initialisation protocol. (Yasso20 differs slightly in how the climate forcing is integrated intra-annually, but the functional form is unchanged; it simply has the freedom to apply separate parameters to each pool group.) Residual differences among models are then attributable to structure rather than to parametric choices.

A key innovation is the *transient initialisation* approach (§8). The analytical steady state is not abandoned but *pushed back in time*: instead of imposing equilibrium at the start of the observation period (1985), each model is initialised at its analytical steady state in 1917 and then run transiently forward to 1985. Over this 68-year pre-run the imprint of the 1917 equilibrium assumption largely dissipates — the fast pools fully relax and only the slow humus pool retains appreciable memory of it — so the calibration begins from a dynamically reconstructed, physically grounded 1985 state rather than an equilibrium one. The strength of the historical anchor is itself a Bayesian-calibrated parameter (σ_{init}), so the steady-state starting point carries a calibrated uncertainty rather than being fixed. This change substantially improves each model's ability to track the observed SOC trajectory.

1.2 Models

Six models span a complexity ladder crossed with two climate-integration choices:

Model	Pools	Free params	Climate ξ	Notes
SP1	1	6	single triplet, Yasso07 form	Pure R
TP2	2	8	single triplet, Yasso07 form	ICBM topology, pure R
TP3	3	10	single triplet, Yasso07 form	ASH cascade, pure R
Yasso07	5	20	single triplet	Fortran core
Yasso15	5	26	pool-group-specific	Fortran core (yasso15.so)
Yasso20	5	26	pool-group-specific, monthly T	Same Fortran as Yasso15

Two comparison axes run through this design. First, a *pool complexity* axis: SP1 → TP2 → TP3 → Yasso07 increases the number of pools from 1 to 5 while maintaining the same Yasso07 climate response, isolating the effect of structural complexity on the calibrated SOC trajectory. Second, a *climate-integration* axis: Yasso07 → Yasso15 → Yasso20 adds pool-group-specific climate responses and finer intra-annual temperature integration, isolating the effect of Yasso-specific climate-coupling refinements.

All models share the same AWEN litter fractionation, observation error model, likelihood form, and transient initialisation protocol. Residual differences in calibrated trajectories are therefore structural.

1.3 Pipeline

The pipeline runs in four sequential stages. Data preparation is documented separately in `Data_work.R`.

Script	Stage	Purpose
`run_{Model}_baseline.R`	0 — Baseline	Forward run at published defaults; no calibration
`run_{Model}_transient_calibration.R`	1 — Calibration	MCMC posterior sampling

Script	Stage	Purpose
`run_{Model}_transient_predictive.R`	2 — Predictive	Posterior predictive trajectories, residuals
`run_residual_analysis.R`	3 — Residuals	Random-forest residual analysis
`run_multimodel_comparison.R`	4 — Comparison	Cross-model metrics and overlay figures

Shared MCMC settings are defined in `calib_config.R`. Prior specifications live in `Prior_specs/{Model}_priors.R` sourced at the top of each calibration script.

---

2 Data

A brief data description is given here; the complete treatment is documented in `Data_work.R`.

2.1 Plots and SOC observations

The calibration dataset covers approximately 350–400 permanent Finnish NFI plots classified as calibration-ready (`calib_ready = TRUE` in `site_raw.csv`). Plots are excluded if they have zero or missing litter inputs, absent organic-layer measurements (OFH), or implied mean residence times exceeding 100 years.

Three SOC measurement campaigns provide the calibration targets:

Campaign	Year(s)	Notes
VMI8	1985–86	First national campaign; mineral SOC depth
Biosoil	2006	European network; same depth protocol
Komeetta	2024	Third Finnish campaign; confirms saturation

With three temporally separated observations per plot, the calibration constrains both the 1985–2006 accumulation phase and the subsequent 2006–2024 near-stability. This separation is important: models that happen to fit the first interval but diverge over the second are penalised.

2.2 Litter inputs

Annual AWEN-fractionated litter inputs (tC ha^{-1} yr^{-1}) come from the plot-level dataset of Tupek et al. (Zenodo, DOI: 10.5281/zenodo.19736499). Across the plot population the mean input is not static: it rises by roughly 40% from the mid-1980s to a peak in the mid-2000s and then plateaus (Figure 5c), with substantial between-plot heterogeneity — growing stands rise as they accumulate biomass, while harvested stands fall. This input trajectory is an important constraint on the initialisation problem, discussed in §8.

2.3 Climate

Gridded daily climate data (temperature, precipitation) span 1961–2025 (`nfi_plot_weather_data_1961_2025.nc`). Annual or monthly summaries are computed per plot according to each model's climate-integration requirements.

3 Models

All six models follow first-order kinetics: each pool C_i decomposes at a rate proportional to its size, modulated by a dimensionless climate modifier $\xi \geq 0$. Litter input J is AWEN-fractionated (acid-, water-, ethanol-, non-soluble) and the fractionation is fixed from the input data.

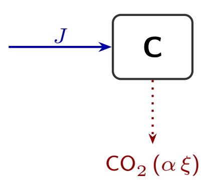
3.1 SP1 (single pool)

All litter inputs — NWL, FWL, and CWL summed across AWEN fractions — enter a single pool C:

$$\frac{dC}{dt} = J(t) - \alpha\,\xi(t)\,C(t).$$

The annual step has a closed-form solution (piecewise-constant J and ξ within each year):

$$C(t+1) = C_{\mathrm{ss}}(t) + \big(C(t) - C_{\mathrm{ss}}(t)\big)\,e^{-\alpha\xi(t)}, \qquad C_{\mathrm{ss}}(t) = \frac{J(t)}{\alpha\,\xi(t)}.$$



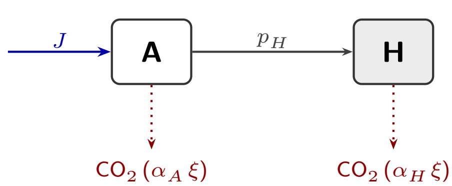
Litter input (blue); respiration (dotted). A single pool decaying at rate $\alpha\,\xi$.

Pure R; no Fortran.

3.2 TP2 (two pools)

Two pools, A (active) and H (humus). All litter enters A; H receives transferred mass from A only:

$$\frac{dA}{dt} = J(t) - \alpha_A \xi(t) A(t), \qquad \frac{dH}{dt} = p_H \alpha_A \xi(t) A(t) - \alpha_H \xi(t) H(t).$$



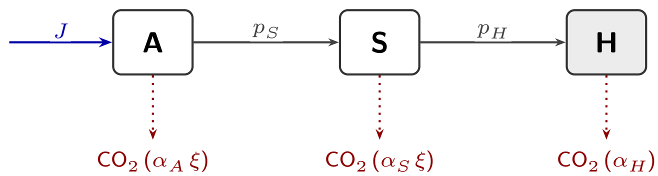
Litter input (blue); humifying transfer $A \to H$ (solid, fraction p_H); respiration (dotted). ξ modulates both decay rates.

This is the ICBM topology (Andrén & Kätterer 1997) with ξ multiplying both pool decay rates. Parameter naming follows Yasso07, making the TP2 parameter set a strict subset of Yasso07. The annual step is solved exactly (see §3.1), not Euler-approximated. Pure R.

3.3 TP3 (three pools)

Three pools in a strict sequential cascade: Active (A) $\to$ Slow (S) $\to$ Humus (H). All litter enters A; a fraction p_S of A decomposition transfers to S; a fraction p_H of S decomposition transfers to H. H has no climate modifier (consistent with TP2 and Yasso conventions). The dynamics are a first-order linear cascade, with J and ξ held constant within each annual step:

$$\frac{dA}{dt} = J - \alpha_A \, \xi \, A, \qquad \frac{dS}{dt} = p_S \, \alpha_A \, \xi \, A - \alpha_S \, \xi \, S, \qquad \frac{dH}{dt} = p_H \, \alpha_S \, \xi \, S - \alpha_H \, H.$$



Sequential cascade $A \to S \to H$ (solid, fractions p_S, p_H); litter input (blue); respiration (dotted). ξ modulates A and S decay; H has no climate modifier.

Each annual step is integrated *exactly*, via the matrix exponential of the cascade generator evaluated in closed form (divided differences of its eigenvalues $-\alpha_A\xi$, $-\alpha_S\xi$, $-\alpha_H$). This is the same analytical treatment SP1 and TP2 apply to their pools, and that the Yasso family applies through its matrix-exponential solver: every model in the ensemble integrates its linear system exactly, so the differences between them are purely structural. (An earlier implementation used an explicit annual Euler step. Being the only approximate integrator in the ensemble, it *inflated* TP3's interannual oscillation with numerical overshoot whenever the calibrated active pool turned over faster than the one-year step; the exact step above removes that artefact, though a smaller physical oscillation genuinely remains — see §13.8.2.) Analytical steady states exist and start the transient pre-run:

$$A^* = \frac{J}{\alpha_A \, \xi}, \qquad S^* = \frac{p_S J}{\alpha_S \, \xi}, \qquad H^* = \frac{p_H p_S J}{\alpha_H}.$$

TP3 sits between TP2 and Yasso07 on the complexity ladder, adding a slow intermediate pool but retaining a sequential (no back-transfer) topology. Pure R.

3.4 The Yasso model family: shared structure

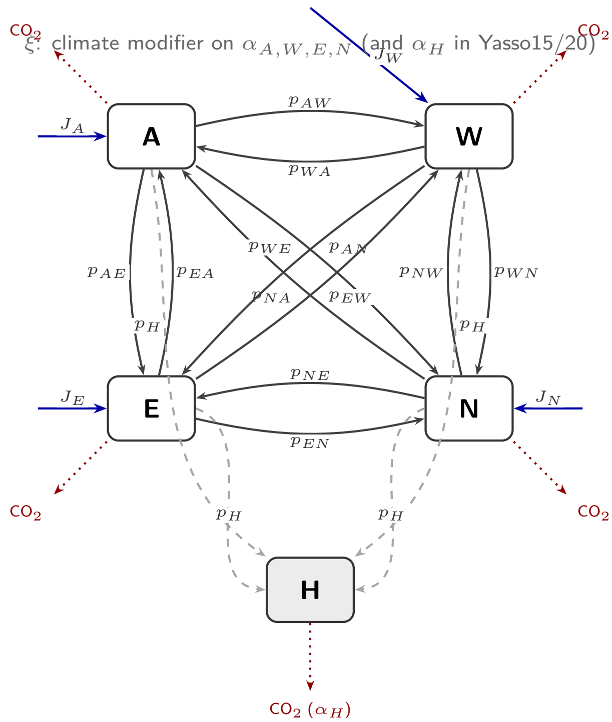
Yasso07, Yasso15, and Yasso20 share the same five-pool AWEN+H topology and the same Fortran matrix-exponential solver. They differ only in how the climate modifier ξ is computed (§3.5–3.7). This subsection describes the shared architecture for reference.

Pool structure and notation

The five pools are: **A** (acid-hydrolysable), **W** (water-soluble), **E** (ethanol-soluble), **N** (non-soluble, largely lignin), and **H** (humus). Litter enters the four AWEN pools in proportions fixed by the chemical fractionation of the input data. Carbon flows between AWEN pools at rates $p_{ij} \cdot \alpha_j \cdot \xi \cdot C_j$, where p_{ij} is the fraction of pool j's decomposition flux routed to pool i. A fixed fraction p_H of each AWEN pool's decomposition flux forms humus; the remainder is respired as CO_2. Humus decomposes irreversibly at rate α_H (no climate modifier in Yasso07; pool-group-specific ξ_H in Yasso15/20).

Pool diagram

The diagram below shows all pools, litter inputs, inter-pool transfers (labelled with HIKET parameter names), humification, and respiration pathways. Litter inputs J_A, J_W, J_E, J_N are AWEN fractions of the annual plot-level litter; for FWL and CWL cohorts a woody-size modifier $h_S(d)$ additionally scales α_j.



Arrow key: →~inter-pool transfer (p_{ij}) - - →~humification (p_H) ⋯→~respiration (CO_2) →~litter input (J)

Parameter summary

Parameter	Symbol	Role
Decomp. rates (AWEN)	$\alpha_A, \alpha_W, \alpha_E, \alpha_N$	Base first-order rate of each AWEN pool (yr^{-1}); fixed at published v
Decomp. rate (H)	α_H	Humus decomposition rate (yr^{-1}); fixed
Transfer fractions	p_{ij}, $i \neq j \in \{A, W, E, N\}$	Fraction of pool j decomp. flux routed to pool i; 12 per model, all f
Humification	p_H	Fraction of each AWEN decomp. flux routed to H; same for all four
AWE climate, linear	β_1	Linear temperature coefficient in ξ_{AWE}; free, log-transformed
AWE climate, quadratic	β_2	Quadratic temperature coefficient; free, unconstrained
AWE precipitation	γ	Precipitation saturation exponent in ξ_{AWE}; free, unconstrained
N-pool climate	$\beta_{N1}, \beta_{N2}, \gamma_N$	As above, for the N pool separately
H-pool climate	$\beta_{H1}, \beta_{H2}, \gamma_H$	As above, for the H pool separately
Size modifier	δ_1, δ_2, r	Reduce α for woody cohorts: $h_S(d) = \min(1, (1 + \delta_1 d + \delta_2 d^2)^{-\lvert r \rvert})$; f
Leaching	$w_1, \ldots, w_5$	Leaching loss weights; inactive (leac = 0) throughout; fixed

Historical litter	σ_{init}	Scales 1917 litter anchor in the pre-run; encodes pre-1985 productiv
Contemporary litter	σ_{input}	Scales observed litter inputs (1985 onward); absorbs litter model un

{Y07 = Yasso07; Y15 = Yasso15; Y20 = Yasso20; Y15/20 = both.}

3.5 Yasso07

Five pools (Tuomi et al. 2009, 2011): A (acid-soluble), W (water-soluble), E (ethanol-soluble), N (non-soluble / lignin), H (humus). Litter inputs enter A, W, E, N proportionally to AWEN fractions; transfers between AWEN pools are controlled by 12 fractions p_{ij}; all four AWEN pools contribute to H at a humification rate p_H; H decomposes irreversibly at rate α_H.

The dynamics are $dC/dt = AC + b(t)$, with the matrix A assembled from decomposition rates and transfer fractions. The Fortran core solves each annual step analytically via the matrix exponential $C(t+1) = A^{-1}(e^{A\Delta t}(AC(t)+b)-b)$, with a Euler fallback for near-singular matrices. NWL, FWL, and CWL litter types are tracked as three parallel cohorts; the per-cohort state vector has 15 elements (5 pools × 3 cohorts).

The climate modifier uses a four-point Gaussian approximation of the intra-annual temperature cycle:

$$\xi(\bar{T}, T_{\text{amp}}, P) = \frac{1}{4}\sum_{k=1}^{4} \exp\big(\beta_1 T_k + \beta_2 T_k^2\big) \times (1 - \exp(\gamma P/1000)),$$

where T_k are four quarterly temperatures derived from annual mean $\bar{T}$ and seasonal amplitude T_{amp}. A woody-size modifier scales decomposition for FWL/CWL cohorts:

$$h_S(d) = \min\Big\{1,\ (1 + \delta_1 d + \delta_2 d^2)^{-|r|}\Big\},$$

where d is the cohort diameter (cm; fixed at 2 cm for FWL, 15 cm for CWL).

3.6 Yasso15

Identical pool topology, AWEN partitioning, and matrix-exponential machinery as Yasso07. The structural change is *pool-group-specific* climate response: three triplets $(\beta_1, \beta_2, \gamma)$, $(\beta_{N1}, \beta_{N2}, \gamma_N)$, $(\beta_{H1}, \beta_{H2}, \gamma_H)$ act on the AWE, N, and H pool groups respectively. The functional form matches Yasso07 (four-point Gaussian temperature approximation, exponential-with-saturation precipitation response). Leaching parameters $w_1, \dots, w_5$ are present in the 35-element parameter vector but are inactive throughout (leaching scalar held at zero).

3.7 Yasso20

Same Fortran core as Yasso15 (compiled as `yasso15.so`, reused). The single structural change is the intra-annual temperature integration: instead of the four-point Gaussian approximation, the modifier is averaged directly over all 12 monthly temperatures:

$$\xi_T^{(g)} = \frac{1}{12}\sum_{m=1}^{12} \exp\big(\beta_1^{(g)} T_m + \beta_2^{(g)} T_m^2\big), \qquad g \in \{\text{AWE}, N, H\}.$$

This is the only computational difference from Yasso15. Monthly climate data (12 rows per year per plot) are required, compared to annual summaries for all other models.

Yasso20 inherits the same 35-parameter structure as Yasso15 in this calibration (all 12 inter-pool flow fractions free; see §5.7).

Note on the published Yasso20 parameterisation. The MAP parameter values from the original global calibration (Viskari et al. 2022) include six transfer fractions constrained to zero. When these published values are applied directly, the per-pool mass budget for pools A and N is marginally exceeded (transfer fractions plus humification fraction sum slightly above 1.0), producing a numerically unstable forward simulation in the HIKET Fortran implementation, which enforces strict diagonal dominance. The HIKET calibration avoids this region of parameter space entirely through the stick-breaking budget constraint (§5.8). The Yasso20 uncalibrated baseline uses FMI prior-centre values, which satisfy mass conservation.

---

4 Fortran implementation: differences from the FMI reference code

The HIKET Fortran cores (`yasso07.f90`, `yasso15.f90`) derive from the FMI Yasso implementations (Järvenpää 2015 for Yasso15/20; Tuomi et al. for Yasso07) but incorporate several modifications introduced in 2025. These are documented here for transparency and to assist cross-comparison with the original FMI code.

4.1 Diagonal dominance check and Euler fallback

The annual timestep in each model is solved via the matrix exponential $\exp(A\Delta t)$. The original FMI code applies the matrix exponential unconditionally. The HIKET implementation adds a pre-check for diagonal dominance: for each column j of A, if the sum of off-diagonal absolute values (outflows to other pools and to humus) exceeds 99.99% of the diagonal absolute value (total decomposition rate), the matrix is classified as near-singular and a first-order Euler step is used instead:

$$C(t + \Delta t) \approx \max(0,\ C(t) + (A\,C(t) + b)\,\Delta t).$$

The condition being detected is mathematically equivalent to the per-pool mass budget $\sum_i p_{ij} + p_H \gtrsim 1$: the sum of all transfer fractions and the humification fraction approaches or exceeds 1, leaving zero or negative respiration from pool j. The $\max(0, \cdot)$ clamp prevents negative pool values in the Euler path but does not conserve mass exactly; repeated Euler steps with clamping can cause slow carbon drift (see §3.7 for the Yasso20 context). The HIKET calibration avoids this regime entirely via the stick-breaking budget constraint.

4.2 Matrix exponential: scaling cap and Taylor terms

The matrix exponential is computed by scaling-and-squaring with a Taylor series expansion. Two changes were made to the original implementation:

- **Scaling-step cap.** The original code had no upper bound on the number of scaling steps j (chosen so that $\|A\| < 2^j$). For ill-conditioned matrices with very large norms, j grew without bound and the squaring loop ran for thousands of iterations, effectively hanging. The cap $j \leq 64$ (covering $\|A\| \leq 2^{64} \approx 1.8 \times 10^{19}$) terminates the scaling phase for any physically meaningful matrix.
- **Taylor terms.** The number of Taylor series terms was increased from 10 to 20. When heavy scaling is applied (large j), the scaled matrix $A/2^j$ is close to zero and many terms are needed for the expansion to accumulate the correction accurately.

4.3 Terminal per-cohort state output (`C_final`)

The Fortran run routines (`yasso07_run_r`, `yasso15_run_r`) return a new output argument `C_final(15)`: the terminal per-cohort pool state $[C_{\mathrm{nwl}} \mid C_{\mathrm{fwl}} \mid C_{\mathrm{cwl}}]$ at the end of the last simulated year. This argument is absent from the original FMI code, which returns only the five summed pool totals.

`C_final` is required for two purposes in HIKET. First, the transient pre-run chains directly into the calibration period: the pre-run's terminal state becomes C_{init} for 1985 (§8). Without the full 15-element per-cohort breakdown, re-initialising the forward simulation would require an analytical steady-state approximation that introduces additional error. Second, the 60-year forward projection (predictive stage) chains from the end of the historical run in the same way. The `C_final` field is attached to the R-side output data frame via `attr(run_out, "C_final")`.

4.4 ξ precomputation in R

The original FMI code computes the climate modifier ξ internally within the Fortran timestep loop, from raw temperature and precipitation inputs. HIKET separates this computation: ξ is computed as an annual (or monthly-aggregated) array in R *before* the Fortran call, and passed to the Fortran as a pre-computed vector argument. This separation has two benefits. First, it allows the Jensen's inequality correction (§8) to be applied consistently and transparently at the R level. Second, it makes the ξ functional form easily modifiable without recompiling Fortran; the different forms for Yasso07 (four-point Gaussian), Yasso15 (same form, pool-group-specific), and Yasso20 (12-monthly average) are handled in dedicated R functions (`compute_xi_yasso07`, `compute_xi_yasso15`, `compute_xi_yasso20`).

4.5 Floating-point precision (Yasso15 and Yasso20)

In `yasso15_mod`, the precision kind is defined as `dp = SELECTED_REAL_KIND(6, 37)`, which evaluates to `REAL(4)` (single precision, $ $7 significant digits) on standard hardware. Despite the `REAL(dp)` declarations throughout the source, all Fortran arguments for Yasso15 and Yasso20 are therefore 4-byte. The R wrappers use `as.single()` for all `.Fortran()` calls to Yasso15/20.

Yasso07 uses `dp = SELECTED_REAL_KIND(15, 307)` = genuine double precision. Its R wrapper correctly uses `as.double()`.

An earlier version of the Yasso15 wrapper incorrectly used `as.double()` for the Yasso15/20 calls. This caused segfaults in the Fortran (type mismatch in the `.Fortran()` interface) that were resolved by reverting to `as.single()`.

4.6 Parameter ordering

The FMI `.dat` posterior files use a different parameter ordering from HIKET. In the FMI files, leaching weights w_1–w_5 occupy positions 17–21, followed by the climate parameters; in HIKET, climate parameters occupy positions 17–25 and leaching weights are at 31–35. The remapping is applied in `Priors_model_matching.R` via the index vector `ORIG_TO_LORENZO` and verified by cross-checking MAP values against named CSV files (`y15par.csv`, `y07par_gui.csv`).

4.7 Yasso07: sign convention for r

The original Yasso07 parameter files store r as a negative value (e.g., -0.307). The Fortran applies $-|r|$ in the size modifier exponent, so the sign is irrelevant at runtime. HIKET stores r as positive and applies a log transform in the calibration (which requires positivity). The `abs()` conversion is applied when reading reference values from `y07par_gui.csv` or `YASSO07_DEFAULT_PARAMS`.

---

5 Parameter specification

5.1 Free, fixed, and derived parameters

For each model, parameters split into:

- **Free**: sampled by MCMC. Includes the two auxiliary uncertainty parameters σ_{init} and σ_{input} in every model.
- **Fixed**: held at published or structurally motivated values. For the Yasso family, decomposition rates and humus parameters are fixed at published defaults, reducing free-parameter count and keeping the intercomparison clean.
- **Derived**: none in the current pipeline.

Transforms map constrained physical parameters to an unconstrained sampling space, with log-Jacobian contributions added to the prior density:

- `log`: positivity; $x = \log\theta$.
- `logit`: $(0, 1)$ bounds; $x = \log(\theta/(1-\theta))$.
- `unconstrained`: identity; no constraint beyond prior.
- `stick_break`: K-fraction simplex constrained to sum $\leq B$; described in §5.8.

5.2 SP1 (6 free, 0 fixed)

Parameter	Meaning	Transform
α	Decomposition rate (yr^{-1})	log

Parameter	Meaning	Transform
β_1	Temperature response, linear	log
β_2	Temperature response, quadratic	unconstrained
γ	Precipitation response	unconstrained
σ_{init}	Historical litter scaling (1917 anchor)	log
σ_{input}	Contemporary litter scaling	log

5.3 TP2 (8 free, 0 fixed)

Parameter	Meaning	Transform
α_A	A-pool decomposition rate (yr^{-1})	log
α_H	H-pool decomposition rate (yr^{-1})	log
p_H	Humification fraction (A $\rightarrow$ H)	logit
β_1, β_2, γ	Climate response	log, unc., unc.
$\sigma_{\text{init}}, \sigma_{\text{input}}$	Litter scalings	log, log

5.4 TP3 (10 free, 0 fixed)

Parameter	Meaning	Transform
α_A	A-pool decomposition rate (yr^{-1})	log
α_S	S-pool decomposition rate (yr^{-1})	log
α_H	H-pool decomposition rate (yr^{-1})	log
p_S	Transfer fraction A $\rightarrow$ S	logit
p_H	Transfer fraction S $\rightarrow$ H	logit
β_1, β_2, γ	Climate response	log, unc., unc.
$\sigma_{\text{init}}, \sigma_{\text{input}}$	Litter scalings	log, log

No fixed parameters. Like SP1 and TP2, TP3 has no published global calibration, so all parameters are learned from the Finnish data.

5.5 Yasso07 (20 free, 6 fixed)

Fixed (6): $\alpha_A, \alpha_W, \alpha_E, \alpha_N$ (AWEN decomposition rates), p_H, α_H. Published Tuomi et al. defaults.

Free (18):

Parameter	Count	Transform
12 transfer fractions p_{ij}, 4 columns of 3	12	stick_break
β_1, β_2, γ	3	log, unc., unc.
δ_1, δ_2, r	3	unc., log, log
$\sigma_{\text{init}}, \sigma_{\text{input}}$	2	log, log

$\delta_2 > 0$ enforced (Fortran NaN for $\delta_2 < 0$ at $d > 0.4$ cm). r stored as positive in HIKET; reference defaults store it negative.

5.6 Yasso15 (26 free, 11 fixed)

Fixed (11): $\alpha_A, \alpha_W, \alpha_E, \alpha_N$, p_H, α_H, $w_1, \ldots, w_5$ (leaching weights; inactive at leac = 0). Published Yasso15 defaults.

Free (28):

Parameter	Count	Transform
12 transfer fractions p_{ij}	12	stick_break
AWE climate: β_1, β_2, γ	3	log, unc., unc.
N climate: $\beta_{N1}, \beta_{N2}, \gamma_N$	3	log, unc., unc.
H climate: $\beta_{H1}, \beta_{H2}, \gamma_H$	3	log, unc., unc.
Size modifier: δ_1, δ_2, r	3	unc., log, log
$\sigma_{\text{init}}, \sigma_{\text{input}}$	2	log, log

5.7 Yasso20 (26 free, 11 fixed)

Identical fixed/free structure to Yasso15. The same 35-parameter vector is used; only the ξ computation method differs at runtime.

Uniform 12-fraction structure across the Yasso family. Earlier Yasso20 calibrations (Viskari 2022) fixed six transfer fractions at zero (structural zeros from a global-dataset MCMC) and derived three algebraically. HIKET does not impose these constraints. All 12 inter-pool fractions are sampled in all three Yasso models, for two reasons. First, the Viskari zeros reflect a warm-site global calibration and are not general biological laws; Finnish boreal data should be free to inform pool-flow structure. Second, fewer free fractions in Yasso20 than in Yasso07 or Yasso15 would confound structural (climate-integration) differences with parametric (degrees of freedom) differences, making the intercomparison uninterpretable. Prior centres for fractions use the published Yasso15 defaults (see §4).

5.8 Stick-breaking parameterisation for transfer fractions

Each column j of the AWEN transfer matrix has three outflow fractions satisfying $\sum_i p_{ij} + p_H \leq 1$ (mass conservation; the unallocated remainder $1 - \sum_i p_{ij} - p_H$ is respired as CO_2). The constraint is enforced via stick-breaking with budget $B = 1 - p_H$. Three unconstrained parameters z_1, z_2, z_3 map to the three fractions by sequentially breaking pieces off the remaining budget:

$$p_1 = B \cdot \sigma(z_1), \quad p_2 = (B - p_1) \cdot \sigma(z_2), \quad p_3 = (B - p_1 - p_2) \cdot \sigma(z_3),$$

where σ is the logistic function. Because each $\sigma(z_i) < 1$, the sum $p_1 + p_2 + p_3$ is strictly less than B, guaranteeing a strictly positive respired fraction. All three fractions in a column are therefore free parameters — none is algebraically determined — and the log-Jacobian of the transform is added to the prior. This guarantees the budget constraint exactly and produces a well-conditioned Jacobian. Each of the four AWEN columns uses an independent stick-breaking transform with the same budget B, contributing $4 \times 3 = 12$ free parameters for the 12 transfer fractions.

5.9 Cross-model parameter overlap

Parameter	SP1	TP2	TP3	Yasso07	Yasso15	Yasso20
α (single rate)	free	—	—	—	—	—
α_A, α_H	—	free	free	fixed	fixed	fixed
α_S	—	—	free	—	—	—
$\alpha_W, \alpha_E, \alpha_N$	—	—	—	fixed	fixed	fixed
p_H	—	free	free	fixed	fixed	fixed
p_S	—	—	free	—	—	—
12 transfer fractions p_{ij}	—	—	—	free	free	free
β_1, β_2, γ	free	free	free	free	free (AWE)	free (AWE)
$\beta_{N1}, \beta_{N2}, \gamma_N$	—	—	—	—	free	free

$\beta_{H1}, \beta_{H2}, \gamma_H$	—	—	—	—	free	free
δ_1, δ_2, r	—	—	—	free	free	free
$w_1, \ldots, w_5$	—	—	—	—	fixed	fixed
$\sigma_{\text{init}}, \sigma_{\text{input}}$	free	free	free	free	free	free
Total free	**6**	**8**	**10**	**20**	**26**	**26**

---

6 Priors

All priors are normal in unconstrained (transformed) space, centred at published model defaults. The joint prior factorises across parameters:

$$\log p(x) = \sum_i \log \mathcal{N}(x_i;\, x_i^{\text{ctr}}, \sigma_{\text{ppm},i}) + \log J(x),$$

where x^{ctr} is the unconstrained image of the physical-space prior centre, σ_{ppm} is the prior standard deviation in unconstrained space, and $\log J$ is the log-Jacobian of the transforms. Prior specifications are stored in `Prior_specs/{Model}_priors.R` and sourced at the top of each calibration script.

6.1 Homogenization across models

The scientific question — whether divergent SOC projections reflect genuine structural differences or calibration artefacts — requires that the calibration setup itself inject no asymmetry that confounds structure with calibration choice. Priors are therefore homogenized in their *construction method, scale, constraints, and degrees of freedom* — **not** in their raw numbers. Identical numbers would in fact be wrong, because the parameters live on model-specific scales; forcing the same number is precisely what produced the Yasso07 β_2 pathology. Each parameter class is assigned to a tier by a single criterion: *was it consistently and freely calibrated across the original models?*

Parameter class	Centre source	Width source	Transform	Consistently calibrated across originals?
Tier 1 climate & size (β's, γ's, δ's, r)	each model's own published MAP	each model's own published posterior/limit (1σ)	log or unconstrained	yes — free in every original ⇒ per-model widths kept
Tier 2 12 transfer fractions	each model's own published structure	*common* logit SD = 0.4 (identical)	logit (stick-break)	no — Viskari 2022 fixed/derived 9 of 12 ⇒ common prior restores symmetry
Tier 3 auxiliary (σ_{init}, σ_{input})	1.0	common log SD = 0.50; σ_{input} additionally bounded to a physical flux window	log; σ_{input} bounded (scaled-logit)	HIKET-only ⇒ identical everywhere

Tier 1 keeps each model's own published uncertainty because coverage is consistent across the original calibrations. Tier 2 overrides the published fractions with a single common prior precisely because they were *not* consistently calibrated — applying the Tier-1 recipe would freeze Yasso20's nine fixed fractions (zero posterior spread) while Yasso15's stay free, re-introducing the very asymmetry the design exists to remove. What legitimately differs between models — Tier-1 centres and widths, and the parameter set itself (Yasso07 has no separate N/H climate response) — is genuine structure, not a calibration choice.

Tier 3 requires one refinement. σ_{init} and σ_{input} were originally treated identically, as weak log-normal nuisances, but they differ in one decisive respect: unlike the decay rates and transfer fractions, litter input carries an *external physical referent* — a boreal forest's total detritus input to soil is bounded by its net primary production. A purely weak prior discards that information, and the likelihood exploits the gap: TP2 and TP3 reach posterior-median effective litter fluxes of 34 and 50 tC/ha/yr, four to six times the most productive boreal forest on record, manufacturing carbon to match the observed stock. We therefore bound the *effective flux* $\sigma_{\text{input}} \times J$ to the widest defensible boreal window, [0.5, 8.7] tC/ha/yr, anchored on the Gower et al. (2001) boreal total-NPP range (52–868 gC,m^{-2},yr^{-1}; at steady state a forest's total litter input $\approx$ its NPP, and NPP is a *generous* ceiling because harvested stemwood leaves as timber rather than litter). The bound is homogeneous across all six models — identical in physical units, not in the abstract multiplier, since J differs slightly between models — and is implemented as a bounded (scaled-logit) reparameterisation rather than a hard truncation, so the sampler works in an unconstrained space and incurs no boundary-rejection artefact; the posterior may still pile up against the wall (the intended diagnostic), but its mixing does not degrade there. σ_{init}, whose posteriors are already physical (0.19–0.72, all below 1 — the post-disturbance recovery signature), keeps only a light guardrail ([0.1, 1.5]). Removing σ_{input}'s escape hatch forces any residual misfit onto the (prior-pinned) rate parameters or into a visible drop in predictive skill, exposing structural inadequacy instead of hiding it — for TP2 and TP3 the inflated multiplier was itself a structural-adequacy signal. The full literature basis, cross-checks against measured Finnish litter, and design decisions are in the accompanying note (`sigma_input_physical_bounds_note`).

6.2 Sources

Model	Source for prior centres	Source for prior widths
SP1	Hand-set (no published global calib.)	Climate on Yasso07 scale; rest weak
TP2	Hand-set ICBM-style defaults	Climate on Yasso07 scale; rest weak
TP3	Hand-set; extends TP2 with Slow pool	Climate on Yasso07 scale; rest weak
Yasso07	`y07par_gui.csv` (Tuomi et al. 2011)	Tuomi 2009 T3 (climate), 2011 T4 (woody)
Yasso15	`Yasso15.dat` posterior mean (FMI)	`Yasso15.dat` posterior SD
Yasso20	`Yasso20_sample_parameters.rda` (FMI)	`Yasso20.dat` posterior SD

The FMI repository is at https://github.com/YASSOmodel/Ryassofortran/tree/master/data. The `.dat` files contain 10,000 MCMC samples from the original calibrations (row 1 = MAP, rows 2–10,001 = posterior draws). Column ordering in the FMI files differs from HIKET's parameter ordering; remapping is performed and verified in `Priors_model_matching.R`.

For Yasso07, no posterior file is publicly available; σ_{ppm} is derived from the published posterior confidence limits — climate parameters from Tuomi et al. (2009) Table 3, woody-size parameters from Tuomi et al. (2011) Table 4. These "±'' limits are read conservatively as 1σ (not divided by 1.96), giving broader, less-informative priors; for log-transformed parameters the relative SD is the limit divided by the paper MAP (delta method). For SP1, TP2, and TP3 the climate widths are placed on this same Yasso07 empirical scale — their climate centres were already borrowed from the Yasso07 MAP — while their remaining structural parameters stay weakly informative so the Finnish data dominates.

This replaces an earlier hand-set $\sigma_{\text{ppm}}(\beta_2) = 0.05$ shared by Yasso07/SP1/TP2/TP3, which was $ 100 – –360×$ looser than the empirical Yasso15/20 widths. Because β_2 multiplies T^2 ($ 600*atborealseasonalextremes*),*thatwidthlettheclimat*
span tens of orders of magnitude, producing 25–46,% non-finite MCMC proposals under the all-or-nothing likelihood. The 1σ value (6.5×10^{-4}) keeps the prior-predictive temperature factor bounded (99th percentile ≈ 9 rather than $\sim 10^{31}$).

6.3 Transfer-fraction priors

For all three Yasso models the prior standard deviation on every transfer fraction is $\sigma_{\text{ppm}} = 0.4$ in unconstrained (stick-breaking logit) space, identical across models. This is the only prior class given a literally identical width everywhere: pool-flow structure is the primary structural-comparison axis, and no single model's published structure should be allowed to dominate — six of Yasso20's fractions were structural zeros in Viskari et al. (2022), and the homogenization deliberately lets the Finnish data, not the original warm-site constraints, decide the flow architecture. The logit scale gives every fraction the same *relative* exploration window regardless of whether its centre is near zero or near one. The chosen width of 0.4 keeps a small fraction (centre 0.01) within $ 2.2×$ of its centre at $+2\sigma$, versus $ 7×$ under the previous $\sigma_{\text{ppm}} = 1.0$; this stops the chains from wandering into forward-model-exploding regions without manufacturing convergence. The fractions remain genuinely non-identifiable (only two SOC observations per plot constrain $ $12 flows), so their R-hat is expected to stay poor — that poor identifiability is a scientific finding, not a defect to be tuned away. Prior centres for the fractions use the published Yasso15 defaults for all three Yasso models (Yasso20 included, since the previously-fixed zeros are now free parameters and Yasso15 defaults are the closest available nonzero prior centres from the same chemical-fractionation framework).

6.4 Climate-response and size-modifier priors

Climate and size parameters receive *more informative* priors than the other parameter classes (though still relatively broad in absolute terms), derived from the FMI posterior (Yasso15/20) or from the Tuomi 2009/2011 published posterior limits (Yasso07). This prevents two pathologies:

1. **The γ,/,σ_{input} compensation loop.** Both parameters scale total decomposition; without an informative prior on γ they trade off freely, producing a ridge in the posterior that destabilises MCMC mixing.
2. **The $\delta_2 < 0$ NaN pathology.** The Fortran size modifier returns NaN for $\delta_2 < 0$ at $d > 0.4$ cm; a log transform with informative prior keeps δ_2 in the well-behaved region.

γ_H for Yasso15 is capped at $\sigma_{\text{ppm}} = 1.5$: the H-pool precipitation response is near-unidentifiable at Finnish precipitation levels (~ 600 mm). The large raw posterior SD (~ 3) reflects a flat likelihood surface, not genuine uncertainty.

6.5 Numerical values

Yasso07

Parameter	x^{ctr} (physical)	σ_{ppm}	Space
p_{ij} (12)	published defaults	0.4	logit (stick-break)
β_1	0.0987	0.26	log
β_2	-0.00157	6.5×10^{-4}	physical
γ	-1.272	0.20	physical
δ_1	-1.708	0.16	physical
δ_2	0.859	0.12	log
r	0.307	0.042	log
$\sigma_{\text{init}}, \sigma_{\text{input}}$	1.00	0.50	log

Yasso15

Parameter	x^{ctr}	σ_{ppm}	Space
p_{ij} (12)	published defaults	0.4	logit (stick-break)
β_1	0.0906	0.0459	log
β_2	-2.15×10^{-4}	1.4×10^{-4}	physical
γ	-1.809	0.0702	physical
β_{N1}	0.0488	0.105	log
β_{N2}	-7.92×10^{-5}	8×10^{-5}	physical
γ_N	-1.173	0.155	physical
β_{H1}	0.0352	0.135	log
β_{H2}	-2.08×10^{-4}	1.6×10^{-4}	physical
γ_H	-12.54	1.50 (capped)	physical

Parameter	x^{ctr}	σ_{ppm}	Space
δ_1	-0.439	0.328	physical
δ_2	1.267	0.286	log
r	0.257	0.055	log
$\sigma_{init}, \sigma_{input}$	1.00	0.50	log

Yasso20

Parameter	x^{ctr}	σ_{ppm}	Space
p_{ij} (12)	Yasso15 defaults	0.4	logit (stick-break)
β_1	0.158	0.104	log
β_2	-0.002	5.4×10^{-4}	physical
γ	-1.440	0.146	physical
β_{N1}	0.170	0.0625	log
β_{N2}	-0.005	4.1×10^{-4}	physical
γ_N	-2.000	0.0353	physical
β_{H1}	0.067	0.0912	log
β_{H2}	0.000	9×10^{-5}	physical
γ_H	-6.900	1.391	physical
δ_1	-2.550	0.435	physical
δ_2	1.240	0.211	log
r	0.250	0.0545	log
$\sigma_{init}, \sigma_{input}$	1.00	0.50	log

SP1, TP2, TP3

Parameter	x^{ctr}	σ_{ppm}	Space
SP1			
α	0.09	0.50	log
β_1, β_2, γ	0.095, -0.00014, -1.21	0.26, 6.5×10^{-4}, 0.20	log, phys., phys.
TP2			
α_A, α_H	0.73, 0.0015	0.50, 0.50	log
p_H	0.028	0.40	logit
β_1, β_2, γ	0.095, -0.00014, -1.21	0.26, 6.5×10^{-4}, 0.20	log, phys., phys.
TP3			
$\alpha_A, \alpha_S, \alpha_H$	0.73, 0.10, 0.0015	0.50, 0.50, 0.50	log
p_S, p_H	0.028, 0.50	0.40, 0.40	logit
β_1, β_2, γ	0.095, -0.00014, -1.21	0.26, 6.5×10^{-4}, 0.20	log, phys., phys.
All three			
$\sigma_{init}, \sigma_{input}$	1.00	0.50	log

---

7 Uncalibrated baselines

Before examining what calibration against the new data tells the models, it is useful to know what the models say *without* any such information — that is, how well published global parameter sets predict the observed SOC stocks. These uncalibrated baseline runs serve as both a diagnostic and a narrative anchor: they show where calibration starts and how far it moves the predictions.

7.1 Purpose and scope

Baseline scripts (`run_yasso07_baseline.R`, `run_yasso15_baseline.R`, `run_yasso20_baseline.R`) run each Yasso model forward at its published default parameters across all calibration-ready plots. No MCMC, no posterior — a single deterministic forward run per plot. Because SP1, TP2, and TP3 have no published global calibration, no baselines are produced for them; the natural reference for the simple models would be their prior centres, but such runs are not currently generated.

Each baseline uses a steady-state initial condition rather than the 68-year pre-run. At published defaults $\sigma_{\text{init}} = 1$ and $\sigma_{\text{input}} = 1$, so the steady state and the pre-run endpoint coincide; the simpler steady-state formulation is used for transparency. The two auxiliary uncertainty parameters are therefore not exercised in the baseline — their effect is, by construction, the identity.

7.2 Parameter choices

The baseline parameter vector for each model combines the published decomposition rates, transfer fractions, climate-response parameters, and size-modifier parameters from the source indicated below. Auxiliary parameters are fixed at $\sigma_{\text{init}} = \sigma_{\text{input}} = 1$.

Model	Transfer fractions	Climate / size parameters
Yasso07	Tuomi et al. published defaults (`y07par_gui.csv`)	Same source
Yasso15	Published Yasso15 defaults (`y15par.csv`)	Same source
Yasso20	Yasso15 published defaults	FMI MAP (`Yasso20_sample_parameters.rda`)

For Yasso07 and Yasso15 the baseline is the full published parameter set. For Yasso20, the climate and size parameters are the FMI MAP values, but the 12 transfer fractions use the Yasso15 published defaults rather than the Viskari (2022) Yasso20 fraction values. At the published Yasso20 fractions, the per-pool outflow for two pools marginally exceeds the available budget; the HIKET solver's diagonal-dominance guard (see the Fortran-implementation section) classifies this as non-conservative and the run is numerically unstable. The Yasso15 fractions satisfy the budget and run stably. This substitution applies only to the Yasso20 baseline; the Yasso20 calibration (§5.7) samples all 12 fractions freely under the budget constraint and is unaffected.

Status. The choice of Yasso20 baseline parameterisation is provisional. An alternative would be to reproduce the original Yasso20 baseline using the unmodified FMI reference code, which tolerates the published fractions without the diagonal-dominance guard. Which option to adopt is left open pending the manuscript narrative.

7.3 Diagnostic outputs

Each baseline produces the following outputs in `diagnostics/{Model}_baseline/`:

File	Content
`*_mean_soc_trajectory.png`	Mean predicted SOC ± 1 SE across plots vs observed campaign means
`*_delta_soc_trajectory.png`	Mean annual ΔSOC rate (tC ha^{-1} yr^{-1}) vs observed
`*_obs_vs_pred.png`	Scatter coloured by vegetation type (`kasvup_tyyppi`)
`*_residuals_diagnostic.png`	6-panel residual diagnostics
`*_rf_importance.png`	Random-forest permutation importance of site covariates
`*_worst25_timeseries.png`	Time series for the 25 worst-fitting plots
`*_residuals_histogram.png`	Distribution of log-scale residuals
`*_residuals.csv`	Per-observation residuals
`*_plot_ranking.csv`	Per-plot mean absolute log-residual
`*_report.txt`	Summary metrics (R^2, RMSE, bias, MAPE) and RF results

The mean absolute SOC and the annual ΔSOC trajectory plots use the same visual form as the calibrated predictive outputs, allowing direct side-by-side comparison of baseline and posterior trajectories in the manuscript figures.

8 Transient initialisation

8.1 Why initialisation matters for the Finnish data

The observed Finnish SOC trajectory — accumulating from 1985 through Biosoil (2006), then approximately stabilising toward Komeetta (2024) — is not easily reproduced by a model pinned to an analytically-imposed steady state with respect to its *1985* litter inputs. Two features of the data shape the initialisation problem. First, litter inputs are not static: the cross-plot mean rises by roughly 40% from the mid-1980s to a peak in the mid-2000s and then plateaus (Figure 5c), though individual plots are heterogeneous — some rise steeply as stands accumulate biomass, others stay flat or fall after harvest. Second, this input rise is of comparable magnitude to the observed SOC accumulation (mean stocks also rise by roughly 40% between 1985 and 2006), and the stocks level off at about the same time the inputs peak.

These two facts have an important consequence. A rising litter supply is itself a candidate driver of the accumulation: even a soil initialised at equilibrium gains carbon while its inputs climb. But a model pinned to $C(1985) = C_{\mathrm{ss}}(1985\text{ litter})$ can express only this one mechanism — the transient response to rising inputs — and must fit the entire accumulation through it, with no freedom to represent a 1985 soil that was already *below* the level its contemporary inputs would sustain. If the true 1985 state was sub-equilibrium — as the management history of Finnish forests suggests, the post-war decades of intensive clearcutting, drainage and fertilisation having left many stands still rebuilding carbon by 1985 — that disequilibrium contribution is instead forced into implausible kinetic parameters, or the model is rejected outright.

The transient initialisation approach removes this restriction. By reconstructing the 1985 state from a calibrated pre-run (§8) rather than fixing it at the contemporary equilibrium, it lets the data apportion the observed accumulation between the two mechanisms: the transient response to rising inputs, and recovery from a sub-equilibrium initial state. The free parameter σ_{init} controls this apportionment: it sets the 1985 stock relative to the contemporary-input equilibrium, so a posterior favouring $\sigma_{\mathrm{init}} < 1$ is direct evidence that the 1985 soil was below equilibrium and that disequilibrium recovery contributes to the observed trend — over and above the accumulation that the rising contemporary inputs alone would produce.

8.2 The 68-year pre-run

Instead of computing $C(1985)$ analytically from contemporary inputs, each MCMC draw runs the model forward 68 years from 1917 to 1985, ending at a calibrated initial state.

The choice of 1917 (Finland's independence) as the pre-run anchor is partly practical (a natural reference point for Finnish forestry history) and partly statistical (68 years is long enough that the fast AWEN pools are approximately at steady state by 1985 regardless of the start condition; only H retains meaningful memory of the initial state).

During the pre-run, two inputs must be specified:

Climate: No gridded climate data exist before 1961. The pre-run therefore uses a constant climate modifier equal to $\xi(\overline{\text{climate}})$ — the modifier evaluated at the mean of the observed climate window (1985–2004). This is an approximation, but an appropriate one: the Yasso climate response is smooth and the mean is a reasonable representative value for this multi-decadal period.

Litter: Litter inputs are linearly interpolated between a historical anchor at 1917 and the start of the observed series at 1985:

$$J(i) = J_{1917} + \frac{i-1}{67}\,(J_{1985} - J_{1917}), \qquad i = 1, \ldots, 68.$$

The anchors are parameterised as:

$$J_{1917} = J_{\mathrm{full\,mean}} \cdot \sigma_{\mathrm{init}} \cdot \sigma_{\mathrm{input}}, \qquad J_{1985} = J_{t_0\,\mathrm{mean}} \cdot \sigma_{\mathrm{input}},$$

where $J_{\mathrm{full\,mean}}$ is the mean litter input over the full observed series and $J_{t_0\,\mathrm{mean}}$ is the mean over the first five observed years (1985–1989). The pre-run begins at the analytical steady state under J_{1917} and constant climate.

8.3 The two auxiliary uncertainty parameters

σ_{init} and σ_{input} are free parameters in every model, both centred at 1.0 with a weakly-informative log-normal prior (unconstrained-space SD = 0.5, a physical 95% interval of approximately [0.37, 2.72] in the interior). σ_{input} carries one additional constraint the rates do not (below).

σ_{input} scales the entire contemporary litter time series (1985 onward). It absorbs uncertainty in the litter input model — plot-level litter estimates carry uncertainty from both the litter model itself and the representation of understorey vegetation, which is not individually measured. σ_{input} also scales J_{1985}, the endpoint of the pre-run interpolation. Because litter input — unlike the decay rates — is externally bounded by forest productivity, the effective flux $\sigma_{\text{input}} \times J$ is constrained to a physical boreal window, [0.5, 8.7] tC/ha/yr, homogeneous across all six models (§6.1); without this bound TP2 and TP3 inflate it four- to six-fold, above the productivity of any boreal forest.

σ_{init} scales only the 1917 litter anchor, J_{1917}. It encodes the ratio of historical (pre-1985) to contemporary mean productivity. The two parameters have strictly non-overlapping roles in the pre-run formula:

> $\sigma_{\text{init}} > 1$: forests were historically *more* productive (old-growth or pre-intensification conditions). The 1985 initial SOC is above what contemporary litter inputs would sustain at steady state. The model predicts declining SOC from 1985.
>
> $\sigma_{\text{init}} < 1$: forests were historically *less* productive (post-disturbance recovery). The 1985 initial SOC is below the contemporary equilibrium. The model predicts accumulating SOC from 1985, consistent with the Finnish observations.
>
> $\sigma_{\text{init}} = 1$: the 1917 litter was equal to the contemporary series mean; the pre-run approaches the same initial state as the analytic steady state.

The posterior on σ_{init} is therefore a direct inference about the pre-1985 productivity history of Finnish forest plots. A posterior that favours $\sigma_{\text{init}} < 1$ across models constitutes model-based evidence for the post-disturbance recovery narrative.

8.4 Jensen's inequality correction

The steady-state climate modifier used to start the pre-run is computed as $\xi(\overline{\text{climate}})$, not as $\overline{\xi(\text{climate}_t)}$. Because ξ is convex in temperature and concave in precipitation, Jensen's inequality gives $\overline{\xi_t} \geq \xi(\overline{\text{climate}})$; the bias is 5–15% under typical Finnish inter-annual variability. Each model wrapper provides a `compute_xi_mean_*` function that averages the climate scalars over the steady-state window (first 20 observed years) and evaluates ξ once at the means, correctly avoiding the average-of-ξ bias.

8.5 Implementation across models

The pre-run protocol is uniform but adapted to each model's solver:

Model	Pre-run implementation
SP1, TP2	Analytical 68-step R loop (exact per-step solution)
TP3	Analytical 68-step R loop (exact per-step matrix exponential)
Yasso07/15/20	68-row litter input fed to the Fortran solver; terminal per-cohort state extracted via `attr(run, "C_final")`

For the Yasso models, the 1917–1985 propagation is advanced by the compiled Fortran solver itself — the same solver used in the forward run, so the pre-run and the forward simulation share identical numerics. This is a consistency choice, not a question of accuracy: Yasso's annual step is analytical (matrix-based), exactly as the exact per-step solution used by the SP1/TP2 R loops. The reason for calling the real solver is faithful per-cohort bookkeeping: the 15-element terminal state (5 pools × 3 litter cohorts) is returned via `C_final`, an output argument added specifically to support this chaining. Without it, only the 5 summed pool totals would be available, which is insufficient to correctly initialise the per-cohort bookkeeping for the forward simulation.

8.6 Likelihood adjustment under transient initialisation

The engine retains a non-transient code path (the original steady-state initialisation), kept only as a fallback; all six production calibrations reported here run with `transient_init = TRUE`. The two paths differ in how σ_{init} enters the likelihood. In the non-transient path, σ_{init} inflated the observation standard deviation at the first observation per plot — a likelihood patch absorbing initial-state uncertainty. Under transient initialisation σ_{init} has already entered the model dynamics (it scales the 1917 litter anchor of the pre-run, §8), so retaining the patch would double-count its effect; the `transient_init = TRUE` flag therefore removes it. The non-transient path is documented here only for completeness — it is legacy code, not part of the current results.

---

9 Likelihood

9.1 Form

Conditional on parameters, observations are modelled as

$$y_{p,t} \sim \mathcal{N}(\hat{C}_{p,t},\ \hat{C}_{p,t} \cdot \sigma_{\text{obs}}),$$

where $y_{p,t}$ is observed SOC (tC ha^{-1}) at plot p, year t, and $\hat{C}_{p,t}$ is predicted total SOC. This is a multiplicative-normal model: the error standard deviation scales with the prediction, so relative observation error is constant across the SOC range.

9.2 Fixed σ_{obs}

σ_{obs} is held at the empirical coefficient of variation of the SOC observations across all calibration plots and years (≈ 0.45–0.50). It is not sampled.

This is a deliberate design choice. If σ_{obs} were sampled, a structurally inadequate model could inflate it to absorb its residual variance, hiding structural inadequacy. By fixing σ_{obs} at the same empirical CV across all six models, structural differences are forced into the residuals — which is exactly where the intercomparison wants them.

9.3 Per-plot log-likelihood

For plot p with observations at $\{t_1, \ldots, t_{n_p}\}$:

$$\log L_p = \sum_{k=1}^{n_p} \log \mathcal{N}(y_{p,t_k};\ \hat{C}_{p,t_k},\ \hat{C}_{p,t_k} \cdot \sigma_{\text{obs}}).$$

Total log-likelihood: $\sum_p \log L_p$ plus the log-Jacobian of parameter transforms. Plots are evaluated in parallel via `mclapply`.

9.4 Rejection logic

Some parameter draws push a model into numerically invalid territory — a non-finite or non-positive predicted stock, or an outright solver failure — where the likelihood cannot be evaluated (the multiplicative-normal density is undefined for $\hat{C} \leq 0$). For these the log-likelihood returns $-\infty$ and the proposal is rejected. This is a robustness guard, not an efficiency device or a tuning knob: it stops invalid proposals (and any NaN/Inf they would propagate) from contaminating the chains, while leaving the posterior shape in the valid region untouched. Three hard rejections (return $-\infty$) apply:

1. $\hat{C}_{p,t}$ is non-finite at any observation year.
2. $\hat{C}_{p,t} \leq 0$ at any observation year (undefined in the multiplicative-error model).
3. The model run returned `NULL` (Fortran crash or wrapper exception).

No physical-ceiling guard is applied. Implausible predictions yield large negative log-likelihoods and are rejected by MCMC naturally. The likelihood evaluates $\hat{C}_{p,t}$ only at observation years; trajectory values between campaigns are computed but not entered into the likelihood.

9.5 Holdout split for independent validation

Plots flagged `is_holdout` in `site_raw.csv` are excluded from the likelihood sum: the calibration is fit only to the non-holdout plots. The held-out plots are carried forward (via the inputs RDS) to the predictive stage, where the prediction metrics — R^2, RMSE, median bias, and the parameter-CI coverage — are computed *separately* for the calibration plots and the held-out plots. The held-out figures are a genuine out-of-sample test: those plots never entered the likelihood, so they check that the calibrated models generalise beyond the plots used to fit them rather than only reproducing in-sample fit. Throughout the results, these independent-validation numbers are kept in their own holdout table (Table 27), distinct from the calibration-sample table (Table 26), and the predictive figures are labelled with the sample they show, so it is always explicit whether a number is in-sample or independent. The split is applied uniformly across all six models, using the same `is_holdout` plot assignment so that every model is validated against the same held-out plots.

---

10 MCMC sampling

The Differential Evolution with snooker update sampler (DEzs) is used, via the `BayesianTools` package (Hartig et al. 2023). DEzs proposes new states using weighted differences between current chain states, then accepts/rejects via the standard Metropolis ratio. It mixes well for moderate-dimensional problems (10–50 parameters) and handles correlated posteriors better than random-walk Metropolis.

10.1 Settings

Shared across all six models, defined in `calib_config.R`:

Setting	Value	Note
Chains	5	Sufficient for Gelman–Rubin $\hat{R}$
Iterations	50,000	Per chain
Burn-in	5,000	10% of iterations
Thinning	None	Full chain retained
Initial state	prior centre x^{ctr}	All chains seeded at same point

10.2 Parallelisation

Two levels of parallelism are available:

- **Per-plot, within-chain:** each likelihood evaluation runs the per-plot inner loop in parallel via `mclapply` with `CORES_PER_CHAIN` workers.
- **Across-chain:** chains can be launched in parallel on a multi-core machine, each running its own per-plot parallelism.

The within-chain mode is mandatory on macOS (nested forking is unsafe).

---

11 Sanity checks and diagnostics

11.1 Pre-MCMC sanity check

Before launching MCMC, each calibration script runs a forward simulation at the prior centre for four randomly selected plots. Three conditions must hold:

1. All predicted SOC values are finite and positive.
2. Predicted-to-observed ratio lies in $[0.1, 10]$ at every observation year.

3. The likelihood `ll_fn(best_x)` is finite across all calibration plots.

A diagnostic PNG is written (`{Model}_forward_at_defaults_{RUN\_ID}.png`). Condition 1 or 3 failure aborts the calibration; condition 2 produces a warning only.

Input diagnostics (`diagnostics_inputs.R`, run separately) confirm that implied mean residence times (MRT = mean SOC / mean litter) do not exceed 100 years. Plots exceeding this threshold are excluded upstream (`calib_ready = FALSE`).

11.2 Convergence diagnostics

Computed on post-burnin samples in unconstrained (sampling) space:

Diagnostic	Threshold	Interpretation
$\hat{R}$ (univariate)	< 1.05	Good per-parameter convergence
$\hat{R}$ (multivariate)	< 1.05	Good joint convergence
Effective sample size	> 200	Adequate independent draws
Acceptance rate	0.2–0.4	Reasonable DEzs mixing

With 20–26 free parameters in the Yasso models and only two to three SOC observations per plot, most transfer fractions are weakly identified by the aggregate SOC likelihood. It is tempting to expect this to show up as poor $\hat{R}$, but that conflates non-identifiability with poor *mixing*: once the sampler mixes cleanly, it traverses the non-identified ridge without difficulty and $\hat{R}$ can be excellent even for parameters the data does not constrain. The genuine signature of non-identifiability is therefore not $\hat{R}$ but (i) marginals pinned to the prior, (ii) near-perfect anti-correlation ridges between fractions leaving the same pool, and (iii) low KL divergence from prior to posterior. This distinction is borne out by the production run (§13) and is developed there.

11.3 Diagnostic outputs

Written to `Calibration_real_data_transient/diagnostics/{Model}/`:

File	Content
`*_traces_*.png`	Trace plots, one panel per parameter
`*_marginals_*.png`	Prior vs posterior KDE (shared x-grid)
`*_rhat_barplot_*.png`	$\hat{R}$ per parameter
`*_kl_divergence_*.png`	KL divergence posterior $\|$ prior
`*_report_*.txt`	Tabular $\hat{R}$, ESS, quantile summaries
`*_forward_at_defaults_*.png`	Pre-MCMC sanity check (4 plots)

Prior and posterior KDE are evaluated on the same x-grid derived from the prior's 99th-percentile range, so tight priors are not artificially compressed.

---

12 Output files

Written to `Calibration_real_data_transient/runs/`:

File	Content	Used by
`{Model}_chains_{RUN\_ID}.rds`	BayesianTools chain objects (unconstrained space)	Re-diagnostics
`{Model}_posterior_{RUN\_ID}.rds`	Post-burnin samples in physical space	Predictive scripts
`{Model}_metadata_{RUN\_ID}.rds`	Run configuration, wallclock, $\hat{R}$, ESS	Reproducibility

File	Content	Used by
`{Model}_inputs_{RUN\_ID}.rds`	Per-plot input structures used in calibration	Predictive scripts

The inputs RDS contains `climate_by_plot`, `inputs_by_plot`, `litter_means`, `obs_meta`, `plot_info`, `plots_real`, `holdout_plots`, `sigma_obs_fixed`, `sigma_ppm`, `STEADY_STATE_YEARS`, and the transient pre-run parameters (`PREINIT_YEAR`, `N_PREINIT_SMOOTH`). It provides exact provenance: predictive scripts use the same per-plot structures the calibration used, regardless of any subsequent updates to `Data_work.R`.

---

13 Results

This section reports the outcome of the homogenised, transient-initialised calibration. All six models were calibrated against the same Finnish dataset (358 plots, 1121 SOC observations), with an independent holdout of 224 observations (897 retained for the random-forest residual analysis) excluded from the likelihood and scored separately. The narrative proceeds from the figures to the numbers and closes with a short interpretation; the emphasis is on *predictive skill of the ensemble*, with convergence and information-gain treated as supporting evidence.

13.1 Uncalibrated baselines

Before calibration, the three Yasso models were run at their published parameter sets with no input scaling ($\sigma_{\text{input}} = 1$), to establish what the literature defaults predict on Finnish data (Table 24). All three explain almost none of the between-plot variance ($R^2 \approx 0.02$) and, more tellingly, all three *under-predict* the observed stocks. The bias worsens monotonically with model complexity: Yasso07 is nearly unbiased (-1.1 tC ha^{-1}), Yasso15 under-predicts by 13 tC ha^{-1}, and Yasso20 by 41 tC ha^{-1} — a gross under-prediction consistent with the per-pool mass-balance issue documented for the published Yasso20 parameter set (§4 and the accompanying mass-balance note).

Table 24: Uncalibrated baseline performance at published defaults (447 plots, 1121 matched observations). Bias is mean predicted − observed; negative = under-prediction.

Model	R^2	RMSE (tC ha^{-1})	Bias (tC ha^{-1})	MAPE
Yasso07	0.021	54.1	-1.1	50.2%
Yasso15	0.026	52.5	-13.0	44.9%
Yasso20	0.023	60.9	-40.8	45.9%

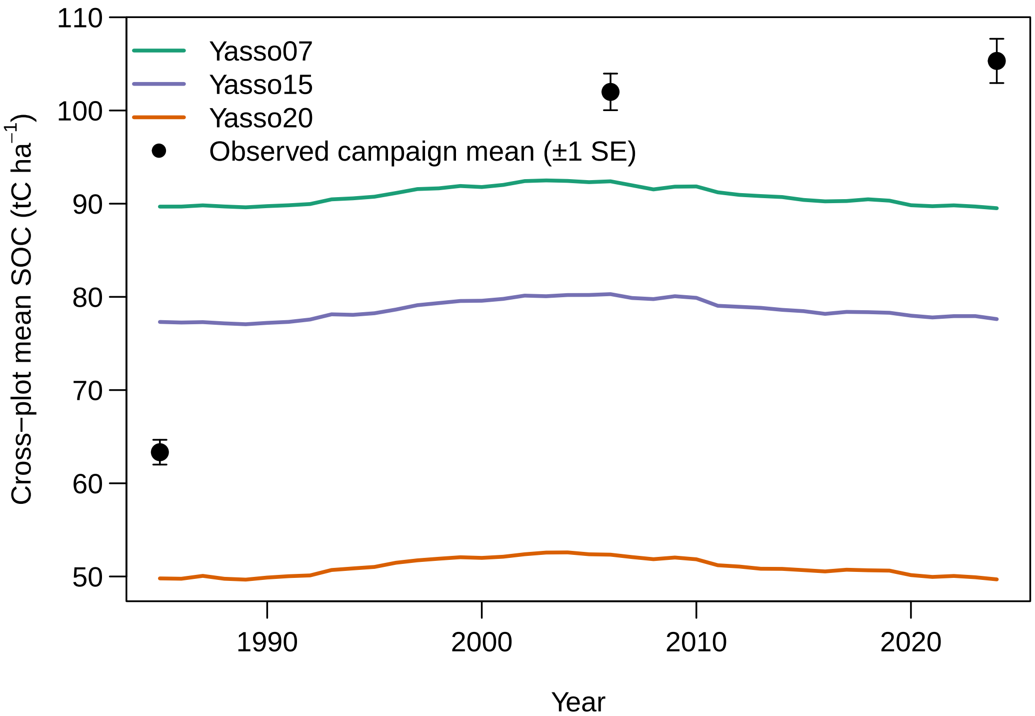


Figure 1: Uncalibrated Yasso baselines at published defaults: cross-plot mean predicted SOC trajectories for Yasso07, Yasso15 and Yasso20 (lines) against the observed campaign means (points, ±1 SE). All three under-predict the 1985–2006 accumulation, and the under-prediction deepens with model complexity — most severely for Yasso20, consistent with the published-parameter mass-balance issue (§4). Yasso20 is shown alongside the other two, not interpreted alone.

13.2 Transient initialisation: evidence for a sub-equilibrium initial state

The transient initialisation (§8) is the keystone methodological choice of the work described here, and the calibration provides a direct test of it. Rather than assuming the 1985 soil sits at steady state, each model reconstructs its 1985 pool state from a 68-year pre-run whose historical litter anchor is scaled by the free parameter σ_{init} — the ratio of 1917 to contemporary litter input. A value below 1 means the soil entered the observation period below equilibrium and still accumulating; a value of 1 recovers a flat, steady-state-like history. Because σ_{init} is estimated rather than fixed, its posterior is itself an answer to the question *does the data want a transient start?*

The posteriors are unambiguous (Figure 2, Table 25): **every model places the entire 95% credible interval of σ_{init} below 1.** Medians range from 0.19 (TP2) to 0.72 (Yasso15), and no model's upper 97.5% bound reaches unity (the highest, Yasso15, stops at 0.91). This is a structure-independent result: regardless of pool count or climate form, the Finnish SOC data reject a steady-state 1985 start and require a sub-equilibrium, accumulating initial condition. The gradient across models is itself informative — the single- and two-pool models infer the most extreme historical deficit (medians near 0.2–0.3), leaning harder on the initial state because they have fewer internal degrees of freedom, whereas the Yasso family, with its slow humified pool, reaches the same observed stocks from a milder deficit (medians near 0.5–0.7). This gradient is the mirror image of the litter-multiplier result (§13.7): the models that infer the largest historical deficit also rescale contemporary inputs the hardest (TP2/TP3 $\sigma_{\text{input}} \approx 13$), because their fast pools store little carbon per unit flux and must be fed harder to reach the observed stocks. So σ_{init} and σ_{input} are not independent findings but two coordinates of a single structural compensation: each model reaches the same SOC by its own trade-off between how far below equilibrium it starts and how much it rescales the flux.

Table 25: Posterior of the historical litter ratio σ_{init} (median and 95% CI). Every interval lies entirely below 1.

Model	Median	95% CI
SP1	0.26	[0.16, 0.37]
TP2	0.19	[0.11, 0.31]
TP3	0.33	[0.19, 0.57]
Yasso07	0.51	[0.40, 0.64]
Yasso15	0.72	[0.59, 0.91]
Yasso20	0.63	[0.48, 0.85]

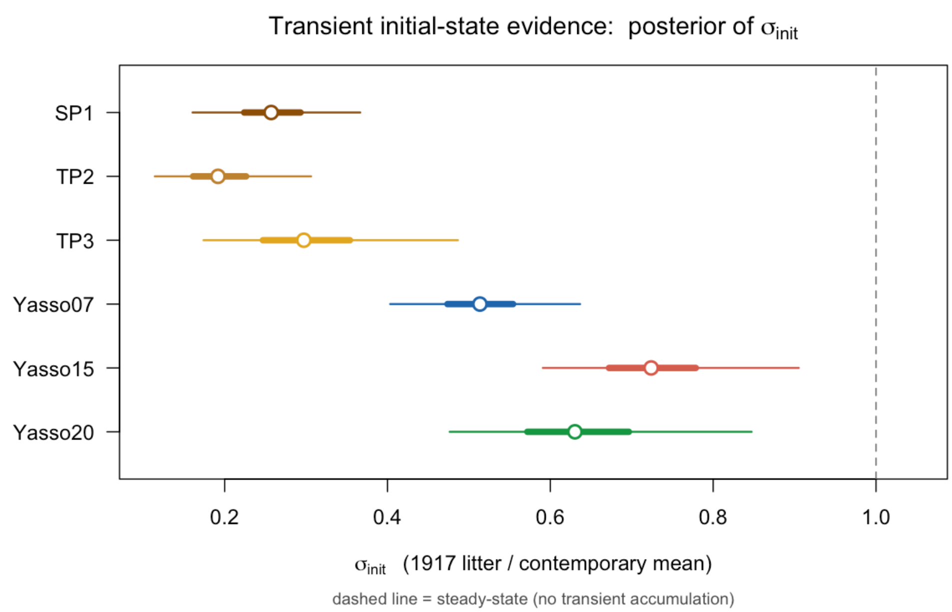


Figure 2: Posterior of the historical litter ratio sigma-init across the six models (open circle = median, thick bar = 50%, thin bar = 95% interval). The dashed line marks the steady-state value of 1; every posterior lies entirely to its left, so all models independently infer a sub-equilibrium, accumulating 1985 state.

This bears directly on HIKET's motivating question. Studies that initialise SOC models at steady state start every model on a flat trajectory, which predisposes them to project saturation or decline; an apparent saturation may then be partly an artefact of the initial condition rather than a genuine structural property of the model. By estimating σ_{init} instead of fixing it, HIKET lets the data decide, and the data decide for accumulation: the same models that would relax to a flat trajectory from a steady-state start instead produce a rising, gradually saturating one once the initial state is freed (Figures 6, 7) — the very behaviour HIKET sets out to study. The models match the 2006 and 2024 campaigns well; the residual is that they begin above the observed 1985 baseline and accumulate somewhat less steeply, which we treat as an open question tied to the coarse, data-free pre-1985 phase (see Discussion) rather than a model deficiency. Ecologically, $\sigma_{\text{init}} < 1$ is consistent with Finnish forest history — lower mid-century litter inputs under younger, less productive, post-war and afforested stands building toward present-day levels — though the parameter conflates that productivity history with any slow disequilibrium and should be read as a magnitude rather than a literal 1917 flux.

13.3 Convergence and sampler health

Sampling was healthy across all six models. Univariate $\hat{R}$ stayed below 1.05 for every parameter in every model with a single exception (SP1 `alpha`, $\hat{R} = 1.092$, discussed below); multivariate $\hat{R}$ ranged 1.007 to 1.067, and effective sample sizes were comfortably above the 200 threshold (ESS 208 to 3663). Invalid-proposal rates were negligible for the well-behaved models (< 0.1%) and elevated only where the posterior has a genuinely heavy tail (SP1 chain 5 12%; TP3 8–17%; both traced to long right tails in `alpha`/kinetic parameters, not to displaced chains).

The single notable $\hat{R}$ signal — SP1 `alpha` — is *not* a non-identifiability symptom but its opposite: `alpha` is the most strongly *identified* parameter in the entire study, with a heavy right tail (its 97.5th percentile is $\sim 6\times$ the median) that is simply hard to sample. Crucially, every Yasso transfer fraction converged in $\hat{R}$ despite being weakly identified by the data. Good $\hat{R}$ is thus the *expected* result of a cleanly mixing sampler and says nothing about whether a parameter is informed; the identifiability question is taken up in §13.7.

13.4 Predictive skill of the calibrated ensemble

The calibrated models are compared in Table 26 (calibration sample) and Table 27 (independent holdout), and visually in Figures 3–4. Three features stand out.

Skill at the level of individual plots is low, and similar across models. Calibration R^2 ranges 0.05 to 0.11 and holdout R^2 0.02 to 0.05; RMSE clusters near 50 tC ha^{-1} (calibration) and 60 tC ha^{-1} (holdout) for all six. This refers specifically to predicting the SOC stock of a *single* plot from its litter and climate — a genuinely hard target: the random-forest residual analysis (§13.7) recovers an out-of-bag R^2 of only 0.34 to 0.52 even with 54 site covariates, so the low ceiling is a property of the data, not of any one model. That none of the models comes close to that ceiling is itself informative — it points to drivers of plot-level SOC that the litter–climate–kinetics structure does not capture, and the residual analysis (§13.7) is where we ask what those drivers might be.

The ranking is consistent, but rests on small absolute differences. TP3 is nominally the best model on both calibration ($R^2 = 0.109$) and holdout ($R^2 = 0.052$) and Yasso20 the weakest ($R^2 = 0.049/0.016$), with the two- and three-pool transient models (TP2, TP3) edging out the full Yasso family. The ordering is stable across the calibration/holdout split — so it is not simply in-sample overfitting — but it should be read with care: at R^2 this low, the models are separated by a few percent of an already small explained-variance budget, and we would not claim any one of them is meaningfully "better' ' at predicting an individual plot. The ranking is useful as a *structural* signal — the simple transient models are at least as good as the full Yasso family on these data — not as a performance verdict.

All calibrated models over-predict, but by structurally different amounts. Median bias is positive for every model, and its size tracks the skill ranking: the simple transient models are closest to unbiased (TP2 6.2, TP3 6.2 tC ha^{-1} in calibration; 2.3 and 1.3 on the holdout), whereas the Yasso family over-predicts more, increasing through Yasso07 (9.3) and Yasso15 (8.8) to Yasso20, the most biased (11.7 tC ha^{-1} calibration, 7.8 holdout). Across all six the calibration bias of 6 to 12 tC ha^{-1} shrinks to 1 to 8 on the holdout — a reversal of the baseline under-prediction. The `Param cov` column (fraction of observations inside the 95% credible interval of the predicted *mean*) is small by construction: this interval reflects parameter uncertainty in the model expectation only and deliberately excludes the multiplicative observation error, so it is a measure of how tightly the mean is pinned down, not a posterior-predictive coverage.

The absolute trajectory reproduces the rising–saturating shape, with a caveat at the 1985 baseline. Plotting mean SOC over time (Figure 7) shows that all six calibrated models reproduce the behaviour at the heart of HIKET: a SOC stock that increases after 1985 and begins to level off toward the recent campaigns — the incipient sink saturation that the Yasso family in particular projects — and they match the 2006 and 2024 campaign means closely. The one clear discrepancy is at the start: the observed mean rises from 63 tC ha^{-1} in 1985 to 102 in 2006 and 105 in 2024 (a +42 increase), whereas every model begins its 1985 trajectory above the observed baseline (predicted 1985 means 76–95) and accumulates somewhat less steeply (+11 to +31). Within this common pattern the ranking is mechanistic: the simpler TP2/TP3 start lowest and accumulate most, tracking the *shape* of the trend best, while the stiffer Yasso family starts highest and accumulates least. We do not read the 1985 offset as a model failure — it is entangled with the deliberately coarse, data-free transient phase and with an unresolved tension in the litter inputs, taken up in the Discussion.

The starting increment encodes the same structural split. The annual-increment view (Figure 6) shows this from a different angle: TP2 and TP3 begin with the steepest accumulation rate, SP1 with the shallowest, and the Yasso family in between. This is the dynamical fingerprint of the input/initial-state compensation above (§13.7). TP2 and TP3 hold carbon in fast pools, so reaching the observed stocks forces an order-of-magnitude input inflation ($\sigma_{\text{input}} \approx 13$) and a deep initial deficit ($\sigma_{\text{init}} \approx 0.2$–$0.3$); starting far below a high, fast-turning equilibrium produces a large early increment that decays as the deficit closes. SP1, with a single slow pool, instead reaches the stocks by *lowering* inputs ($\sigma_{\text{input}} \approx 0.3$) and turning over slowly: its target equilibrium is low and its approach gradual, so the increment is shallow and nearly flat. The spread in starting increments is therefore not noise but the same structural disagreement — about *how* each model reaches the stocks — seen in the time derivative.

The forcing is strongly oscillatory in climate but smooth in litter. Figure 5 summarises the three model drivers as cross-plot means over the observed period. Mean annual temperature and precipitation swing sharply from year to year ($\pm$1–2°C and $\pm$100–150 mm), whereas the litter input rises smoothly to a mid-2000s peak and then plateaus, with no high-frequency component, so any interannual variation in SOC could only come from the climate modifier $\xi(T, P)$, not the inputs. TP3 shows a visible year-to-year oscillation in mean SOC (Figure 7) that the other models do not. An earlier TP3 run inflated it with a numerical artefact — an explicit Euler integrator overshooting a sub-annual active pool — which we removed by integrating TP3 exactly like the rest of the ensemble; but a bounded, *physical* oscillation survives exact integration, because TP3's calibrated active pool genuinely turns over faster than a year and so tracks the climate modifier. The distinction, and why TP2 stays smooth despite an equally fast pool, is developed in §13.8.2.

Table 26: Calibrated predictive performance (full calibration sample, 1121 obs, median-prediction metrics).

Model	R^2	RMSE	Bias	Param cov
SP1	0.065	51.3	+8.2	0.056
TP2	0.089	49.6	+6.2	0.086
TP3	**0.109**	50.0	+6.2	0.089
Yasso07	0.067	52.3	+9.3	0.062
Yasso15	0.060	52.2	+8.8	0.062
Yasso20	0.049	53.6	+11.7	0.054

Table 27: Calibrated predictive performance on the independent holdout (224 obs).

Model	R^2	RMSE	Bias	Param cov
SP1	0.027	60.4	+3.8	0.071
TP2	0.036	59.1	+2.3	0.071
TP3	**0.052**	58.4	+1.3	0.098
Yasso07	0.027	61.9	+5.5	0.062
Yasso15	0.026	60.5	+4.3	0.080
Yasso20	0.016	62.3	+7.8	0.062

13.5 Baseline versus calibrated

Calibration changes the picture in three ways. First, it removes the structural under-prediction: the strongly negative baseline biases (Yasso15 −13, Yasso20 −41 tC ha^{-1}) become small positive biases (+8.8, +11.7). Second, it lifts R^2 from ~ 0.02 to 0.05 to 0.11 — a real but modest gain, reinforcing that the explainable between-plot variance is limited. Third, and most informative, the correction is achieved almost entirely through the litter-input multiplier σ_{input} rather than through the decomposition kinetics, which sets up the identifiability discussion below.

13.6 Residual structure

Beyond the headline metrics, the residuals carry two further signals.

The over-prediction is a uniform multiplicative offset, not a reporting artefact. Re-scoring the calibrated predictions on the log scale — the natural scale of the multiplicative error model — gives a mean log-residual ($\log \text{obs} - \log \widehat{\text{SOC}}$) of −0.10 to −0.16 across the six models, i.e. predictions sit $\approx$ 10–16% high in geometric-mean terms. The value is essentially identical whether the predictive mean or median is used (the predictive spread is narrow), so the bias is not an artefact of summarising a skewed predictive distribution by its mean. It also tracks predictive skill: the offset is smallest for the highest-R^2 models (TP2, TP3, ≈ -0.10) and largest for the lowest (Yasso20, −0.16). We argue in the Discussion (§13.8.1) that this is a property of the proportional likelihood rather than a physical statement about Finnish soils.

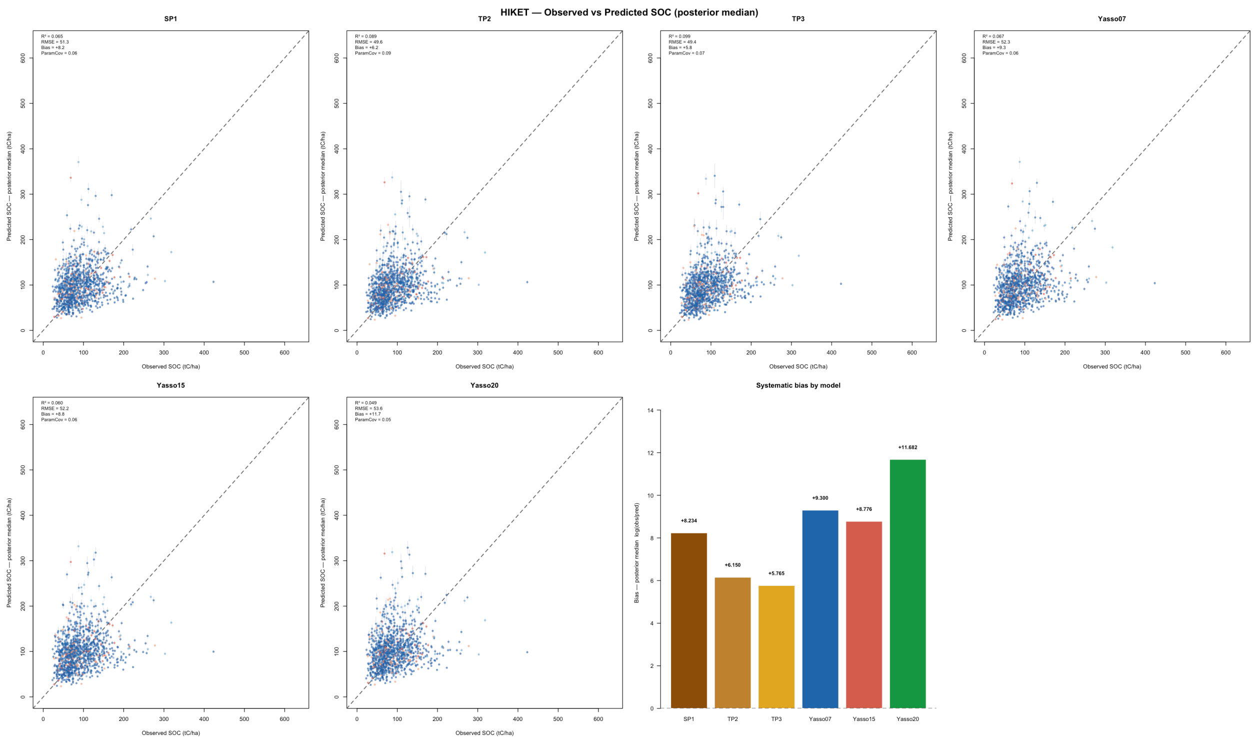


Figure 3: Observed vs predicted SOC across the six calibrated models (calibration sample). Points are plot-level posterior medians; the spread of predictive skill between models is small.

Residuals are site-structured, and the structure is shared across models. A random forest regressing the log-residuals on 54 site covariates recovers an out-of-bag R^2 of 0.34 to 0.52 (Table 28), so a substantial and partly reducible share of the unexplained between-plot variance is site-driven rather than random. The importance ranking is strikingly consistent across all six models (Figure 8): **stand basal area at 1985** (`basal_area_85`) is the single most important predictor in five of the six models — and second in TP3 — followed by soil chemistry (total nitrogen, cation-exchange capacity, exchangeable aluminium, clay) and then the climate variables (mean and warmest-month temperature). No single covariate exceeds ~ 5–9% of total importance, so the residual signal is diffuse rather than concentrated in one driver. Notably, the covariate set *does* include the categorical site classifiers one might expect to carry a regular bias — NFI development class, Cajander fertility class, growing-site type (`kasvup_tyyppi`), and dominant species — yet none of them ranks among the leading predictors (development class, for instance, is 17th–38th). There is thus no clean class-wise residual bias, e.g. by site or soil type, of the kind that might have been anticipated; the residual structure is carried by continuous gradients (biomass, soil chemistry, climate), not by discrete site categories. That structurally different models have their residuals ordered by the *same* covariates confirms the unexplained variance is a property of the data and the site, not of any one model's kinetics.

Table 28: Out-of-bag R^2 of the random-forest residual model, per calibrated model (calibration sample, 54 site covariates, log-residual target).

Model	SP1	TP2	TP3	Yasso07	Yasso15	Yasso20
OOB R^2	0.42	0.50	0.52	0.42	0.38	0.34

13.6.1 [TO DISCUSS — stand structure/age as the leading residual driver]

For joint interpretation. The leading residual predictor is stand basal area at 1985, a proxy for stand development stage and standing biomass; the explicit NFI development class (`dev_class_85`) is by contrast a weak predictor (ranked 17th–38th across models). No direct stand-age variable is currently in the covariate set. Whether to read `basal_area_85` as a stand-age/development signal — and whether to add an explicit age covariate to sharpen it — is the open interpretive question to settle before finalising this paragraph.

13.7 Identifiability and information gain

The convergence diagnostics confirm that good $\hat{R}$ coexists with weak identification. Three independent signatures locate where the data actually informs the parameters.

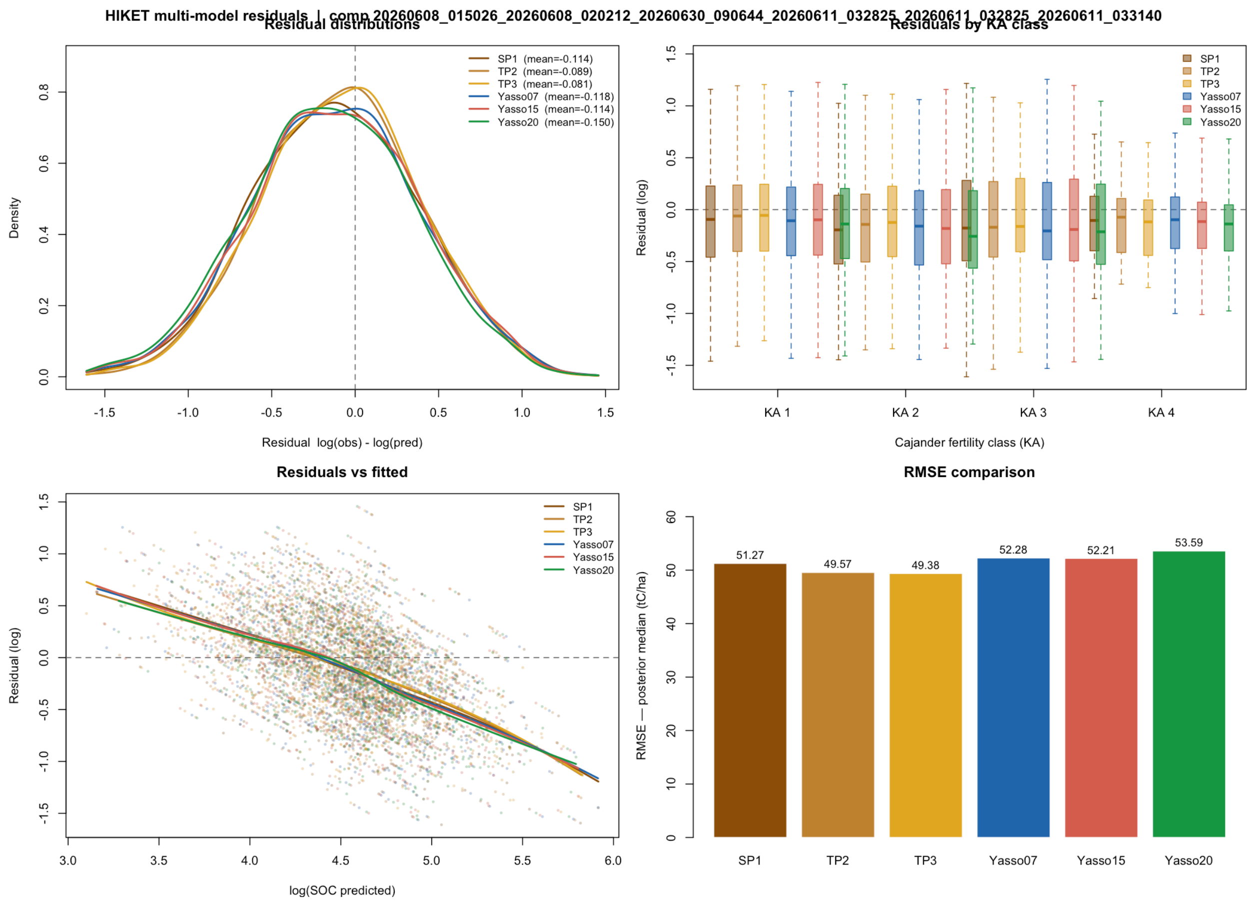


Figure 4: Residual structure across models (calibration sample). Residuals are broadly comparable in scale and share the same site-driven structure (Section 13.7).

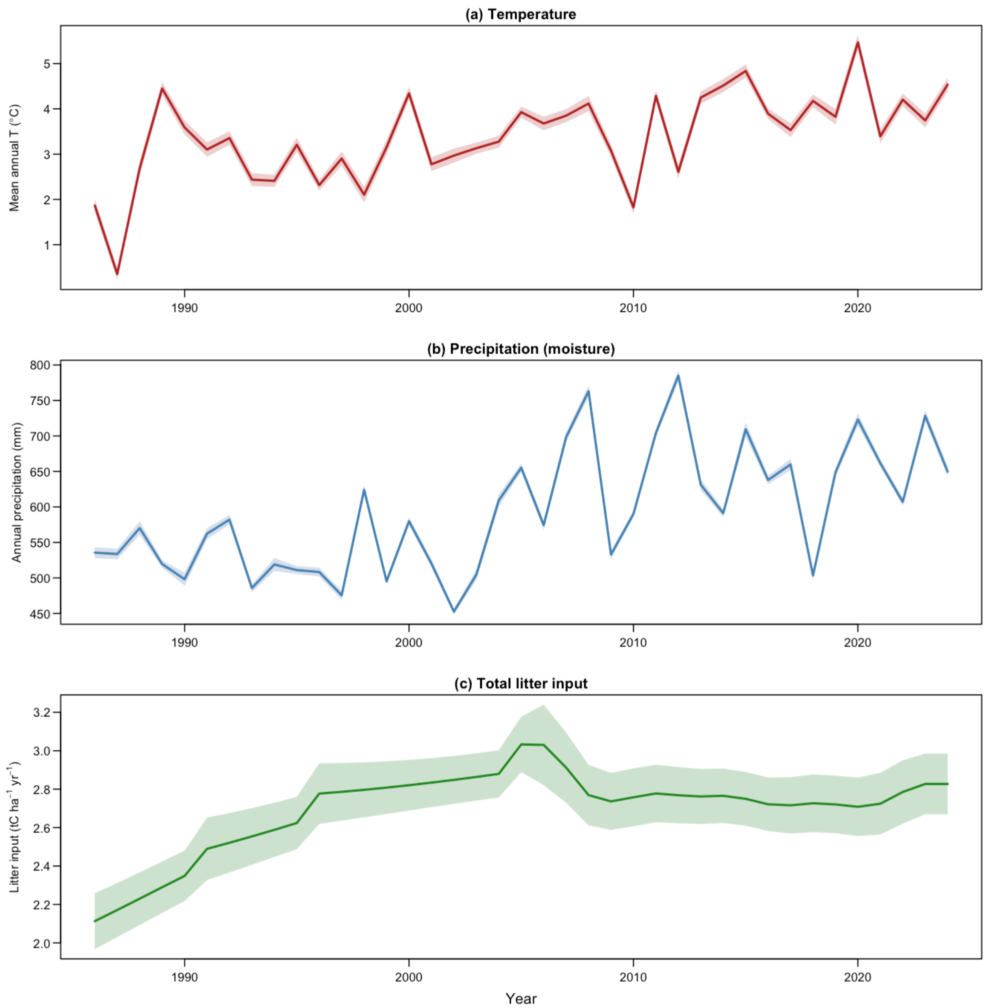


Figure 5: The three model drivers over the observed period (cross-plot mean line, band = 95% CI of the mean; the 1985 initialisation year is omitted). Mean annual temperature (a) and precipitation (b) are strongly oscillatory year-to-year, whereas total litter input (c) rises smoothly and then plateaus. Interannual oscillations in a model's SOC trajectory (Figure 7) therefore reflect the climate modifier, not the inputs.

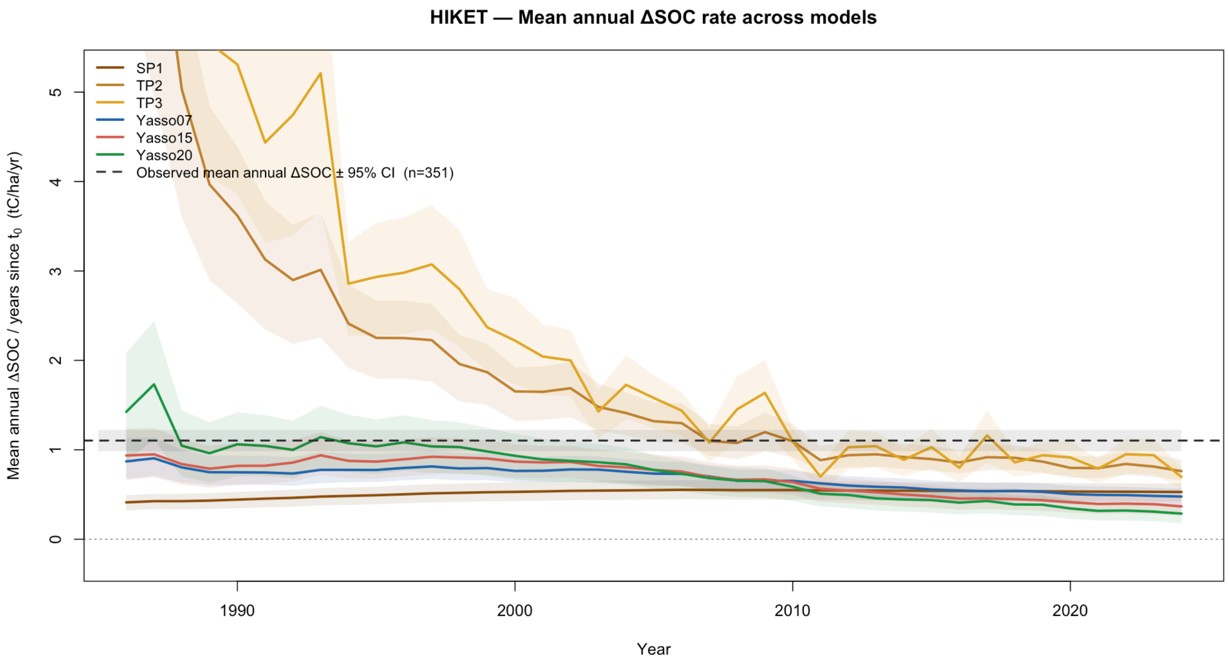


Figure 6: Mean annual SOC increment across models over the projection window. Transient initialisation yields a rising, gradually saturating trajectory; the modelled increment is somewhat lower than the observed accumulation rate (see the absolute levels in Figure 7).

Table 29: Posterior of the litter-input multiplier σ_{input} (median and 95% CI). A value of 1 means no rescaling.

Model	Median	95% CI	Implied input change
SP1	0.31	[0.25, 0.37]	$\sim 3\times$ lower
TP2	13.4	[8.0, 21.1]	$\sim 13\times$ higher
TP3	12.7	[10.3, 15.2]	$\sim 13\times$ higher
Yasso07	1.37	[1.04, 1.74]	near baseline
Yasso15	1.91	[1.58, 2.28]	$\sim 2\times$ higher
Yasso20	2.98	[2.26, 3.91]	$\sim 3\times$ higher

13.8 Discussion

The results support a coherent reading. Once initialisation, likelihood, error model, and priors are homogenised, the six models converge to a narrow band of modest predictive skill, and the small differences that remain track structure coherently across the calibration/holdout split. Crucially, the transient initialisation lets the calibrated models reproduce the phenomenon HIKET set out to examine: a post-1985 SOC increase that begins to saturate toward the present (Figures 6, 7), in place of the flat or declining trajectory a steady-state start imposes. The match at the 2006 and 2024 campaigns is good; the main residual is an over-predicted 1985 baseline and a somewhat shallower accumulation slope, captured best by the simpler models.

That residual is better read as an open question than as a model deficiency, because the pre-1985 phase of the approach is deliberately coarse. No historical litter or climate record exists for 1917–1985: the pre-run interpolates litter linearly from a single calibrated 1917 anchor (σ_{init}) under a constant climate, so the early trajectory rests entirely on calibration and carries large uncertainty. A connected and still-unresolved tension concerns the litter inputs. The plot-level Tupek inputs used here are approximately flat over the observed period, whereas national-scale input reconstructions are lower in the past and rise toward the present — a shape that mirrors the SOC trend itself. If the true historical inputs were lower and increasing, both the over-predicted 1985 baseline and the too-shallow slope would shrink; if the flat plot-level inputs are correct, then the coincidence of a rising SOC trend with flat inputs is itself informative, and may be what masked the initialisation problem in earlier work. The present data cannot settle this. The transient calibration is therefore best framed as a proof of concept — a first demonstration that the approach can represent the observed accumulation and incipient saturation — whose natural successor is a dedicated historical reconstruction supplying genuine priors for the pre-1985 phase rather than leaving it to calibration.

Two structural messages emerge. First, the three simple transient models are not just a low-complexity baseline for the

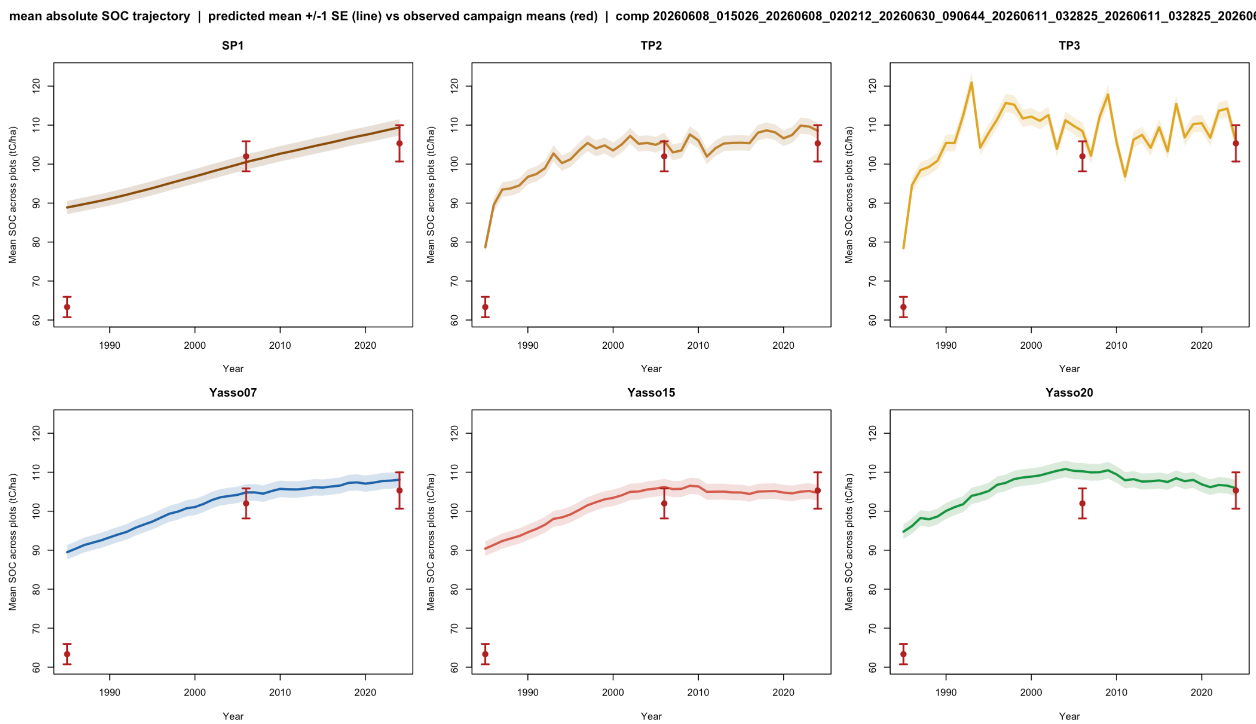


Figure 7: Mean absolute SOC across plots over time for each calibrated model (line = predicted cross-plot mean, band = +/-1 SE; red = observed campaign means +/-95% CI); the calibrated analogue of the published-defaults baseline trajectories. All models reproduce the rising-then-levelling shape and match the 2006 and 2024 campaigns closely, but begin above the observed 1985 baseline and accumulate somewhat less steeply; TP2/TP3 track the shape best, the Yasso family least.

Figure 8: Random-forest residual-predictor importance (relative, within-model) for the calibration sample; the most important covariates are shown and column labels give the out-of-bag R^2. Stand basal area at 1985 (`basal_area_85`) tops every model, followed by soil chemistry and climate, and the ordering is shared across structurally different models.

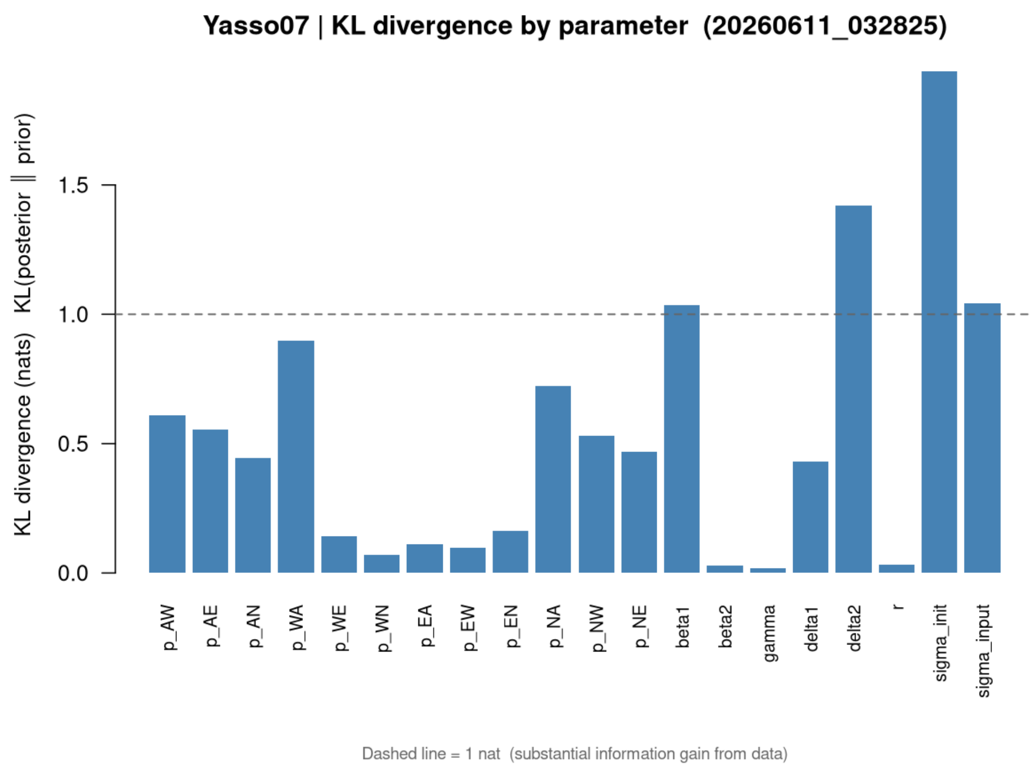


Figure 9: Information gain (KL divergence, posterior ‖ prior) per parameter for Yasso07, bars coloured by parameter class. The 12 transfer fractions (orange) each gain well under 1 nat, while the auxiliary turnover parameters (purple) and climate-and-size modifiers (green) gain the most — the data inform aggregate turnover, not the inter-pool splits.

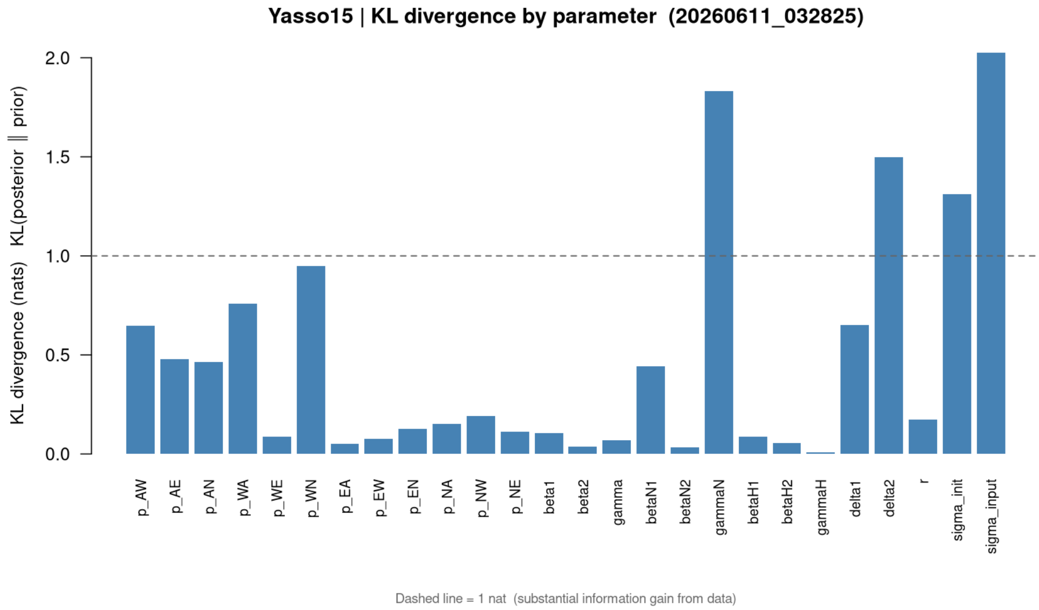


Figure 10: KL divergence per parameter for Yasso15 (same colour scheme). The transfer fractions again gain almost no information relative to the prior.

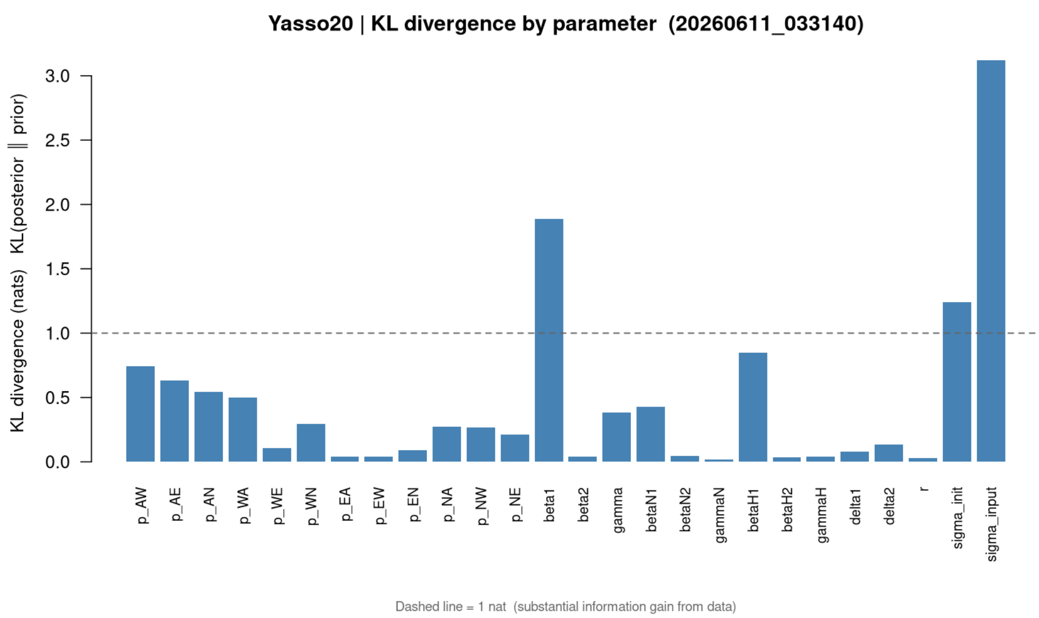


Figure 11: KL divergence per parameter for Yasso20 (same colour scheme). Despite all twelve fractions being free, none is appreciably informed by the data.

remove essentially all of it. Crucially, the bias is the *only* headline sensitive to the likelihood choice: the skill ranking, the fraction non-identifiability, and the input-dominance result are governed by the information content of two SOC observations per plot, not by the error-model shape, and are invariant to it. We therefore report the bias as a modelling-choice consequence and read the structural conclusions as robust to it.

13.8.2 Forcing sensitivity and the resolved TP3 oscillation

Earlier TP3 runs (and, mildly, TP2) showed an interannual ringing in mean SOC, absent in SP1 and the Yasso family. Two distinct effects are superimposed here, and separating them matters. The first was a genuine *numerical* artefact. TP3 was the *only* model integrated by an explicit annual Euler step, and when a pool turns over faster than the step the Euler update factor $(1-\alpha_A\xi)$ turns negative, so the pool overshoots its equilibrium each year. At the posterior median this inflated the trajectory roughness by roughly one to three orders of magnitude (mean $|\Delta^2\mathrm{SOC}|$ up to $\sim 10^3$–10^4 tC,ha^{-1} on individual plots). Replacing TP3's integrator with the exact matrix-exponential step now used throughout the ensemble (§3.3) removes that artefact and cuts trajectory roughness by roughly 5–700×; under constant forcing the exact step relaxes monotonically to the equilibrium the Euler step overshoots.

The exact integrator does *not*, however, make the TP3 trajectory smooth, and it should not. A bounded interannual oscillation remains, and it is *physical.* TP3 reaches the observed stocks not by slow turnover but by holding carbon in a fast active pool fed by heavily inflated inputs ($\sigma_{\text{input}} \approx 13$–$20$; §13.7), which pushes the active pool's residence time below one year: at the posterior median $\alpha_A\xi$ has median ≈ 1.4–2.0 and exceeds~1 in $\sim$95–100% of years, more in warm years. A pool that turns over faster than the annual step re-equilibrates to each year's climate ($A_{\text{ss}} = J/\alpha_A\xi$, small in warm years, large in cold), so total SOC genuinely tracks interannual climate. This residual oscillation is *not* removable by finer time resolution: re-integrating the same posterior with twelve monthly sub-steps — seasonal temperature reconstructed from $\bar{T}$ and its amplitude — reduces roughness by only $\sim$10–20% and shifts mean SOC by < 2 tC,ha^{-1}, because the seasonal cycle is already inside ξ (the four-node quadrature of §3.5) and the residual signal is year-to-year, not sub-annual.

The contrast with the other models is therefore *structural*, not numerical — all six now integrate exactly. It is governed by two quantities: how fast the active pool turns over ($\alpha_A\xi$ relative to~1) and what fraction of the total stock that pool holds. TP2's active pool is comparably fast ($\alpha_A\xi \approx 1.6$) but holds only $\sim$23% of total SOC, the rest buffered in humus, so its total barely ripples (roughness ~ 3.7 vs TP3's ~ 9.3); TP3's fast pool holds a comparable share ($\sim$26%) but its three-pool cascade lets the fluctuation propagate through the intermediate pool as well, and its higher stocks amplify the absolute swing. SP1 sits at the opposite extreme: a single pool turning over on century timescales ($\alpha\xi \approx 0.003$) that simply accumulates litter — perfectly smooth and still rising through the window. The Yasso family parks most carbon in the slow humified pool, so despite fast AWEN pools it never enters the sub-annual stock regime and stays smooth. The genuine structural point is thus unchanged: TP3's fast-pool, inflated-input solution is the same under-identification seen statically as input-dominated information gain and as the elevated invalid-proposal rate (8–17%); the interannual oscillation is a *symptom* of that solution, correctly rendered, not a numerical defect. The mean-SOC and increment figures below are the exact-integrator run; the TP3 oscillation they show is the physical residual, not the former Euler ringing.

13.8.3 [PLACEHOLDER — residual Yasso-specific over-accumulation]

To resolve. After removing the likelihood-induced offset (§13.8.1), test via a symmetric-lognormal refit whether any Yasso-specific over-prediction survives — e.g. transient over-filling of the slow humified pool during the 1917–1985 pre-run. Current evidence bounds it to small.

---

14 Key design decisions

1. **Transient initialisation via a 68-year pre-run.** The 1985 SOC trajectory is inconsistent with a steady-state initial condition under flat litter inputs. The pre-run from 1917 gives all models a calibrated, physically-grounded initial state. σ_{init} is the key parameter encoding pre-1985 productivity history; its posterior is directly interpretable as evidence for or against a post-disturbance recovery narrative.
2. **Uniform pre-run protocol and exact integration across all six models.** SP1/TP2/TP3 use exact analytical per-step solutions (closed form for one and two pools; closed-form matrix exponential for the three-pool cascade) and Yasso07/15/20 use the Fortran matrix-exponential solver with `C_final` chaining. Every model thus integrates its linear pool system exactly, so the inter-model differences are structural, not numerical; and the pre-run produces the same conceptual quantity (initial state at 1985) for all.

3. **Six models spanning two design axes.** Pool complexity (SP1 → TP2 → TP3 → Yasso07) and climate integration (Yasso07 → Yasso15 → Yasso20) vary independently. Conclusions about the saturation signal are drawn from the ensemble, not from individual models.
4. **Multiplicative-normal likelihood.** The standard deviation of the observation error scales with the prediction, representing proportional rather than absolute measurement uncertainty. This is appropriate for SOC stocks, where measurement noise and spatial representativeness both scale with stock magnitude.
5. **Fixed σ_{obs}** at the empirical CV of the SOC observations, identical across all six models. Prevents structurally inadequate models from absorbing their residual variance into the error term; forces differences into the residuals where the intercomparison can examine them.
6. $\xi(\bar{\text{climate}})$ **rather than** $\overline{\xi(\text{climate}_t)}$ for the pre-run climate. Jensen's inequality correction; 5–15% bias in boreal conditions otherwise. Implemented uniformly via `compute_xi_mean_*` functions.
7. **Steady-state climate window = first 20 observed years.** The 1985–2004 climate is used to compute the mean ξ for the pre-run. This pre-dates the accelerated boreal warming of the 2010s and is representative of the pre-run period.
8. **Decomposition rates fixed at published defaults** for the Yasso family ($\alpha_A, \alpha_W, \alpha_E, \alpha_N, p_H, \alpha_H$). Reduces free-parameter count and keeps the intercomparison clean. SP1, TP2, and TP3 have no published global calibrations; all their parameters are free.
9. **All 12 transfer fractions free in all three Yasso models.** The Viskari (2022) structural zeros and derived fractions in the original Yasso20 are not imposed, because they would give Yasso20 fewer degrees of freedom than Yasso07/15 and make the structural intercomparison uninterpretable.
10. **Stick-breaking parameterisation with budget $B = 1 - p_H$.** Guarantees the per-column mass-budget constraint exactly; the remainder of each pool's outflow is respired as CO_2.
11. **Common weakly informative prior on transfer fractions** ($\sigma_{\text{ppm}} = 0.4$, identical across models, in unconstrained logit space). Finnish data drives pool-flow structure — the primary structural-comparison axis — and an identical width removes any model-to-model asymmetry in how freely the flows are explored.
12. **Informative priors on climate and size parameters** (Yasso15/20 from FMI posterior SDs; Yasso07 from the Tuomi 2009/2011 published posterior limits, read as 1σ). Prevents the γ,/,σ_{input} compensation loop and the δ_2 NaN pathology, and removes the β_2 climate-modifier detonation that the earlier hand-set width caused.
13. **Three SOC observation years (1985, 2006, 2024).** The Komeetta campaign (2024) provides a third calibration target. The 1985–2006 interval constrains the accumulation dynamics; the 2006–2024 near-stability constrains long-term decomposition rates and the saturation behaviour. Full Komeetta data description in the data documentation.
14. **SP1, TP2, and TP3 share the Yasso07 ξ formula.** Climate response comparison across the pool-complexity axis is not confounded by approximation-scheme differences.
15. **ICBM topology for TP2.** All litter enters A; H receives only transferred mass. Eliminates a weakly-constrained input-partitioning parameter and is standard in Nordic forestry SOC models.
16. **Sequential cascade for TP3.** No back-transfers between pools; A → S → H only. This is the minimal three-pool generalisation of TP2, extending the complexity ladder by one step before the full AWEN topology of Yasso07.
17. **Global σ_{input}.** A single multiplicative litter scaling per model run. Per-species or per-region stratification is a possible extension but is not used in the current pipeline.
18. **No physical-ceiling rejection** in the likelihood. Implausible predictions are penalised via the log-likelihood, not a hard cap. Consistent with Viskari (2022).

---

15 Appendix: prior and posterior marginals

For every free parameter in every model, the panels below overlay the posterior (solid) on its prior (dashed), both in physical (back-transformed) units. They are the visual counterpart of the identifiability discussion (§13.7): where the data inform a parameter, the posterior contracts and shifts away from the prior; where they do not, the posterior sits on top of the prior. The transfer fractions of the Yasso models are the clearest example of the latter — posterior ≈ prior, the signature of non-identifiability — while the auxiliary turnover and climate parameters show the strongest contraction. Each panel's x-axis is clipped to the union of the posterior and prior central ranges so that heavy prior tails do not compress the posterior to an unreadable spike.

15.1 SP1

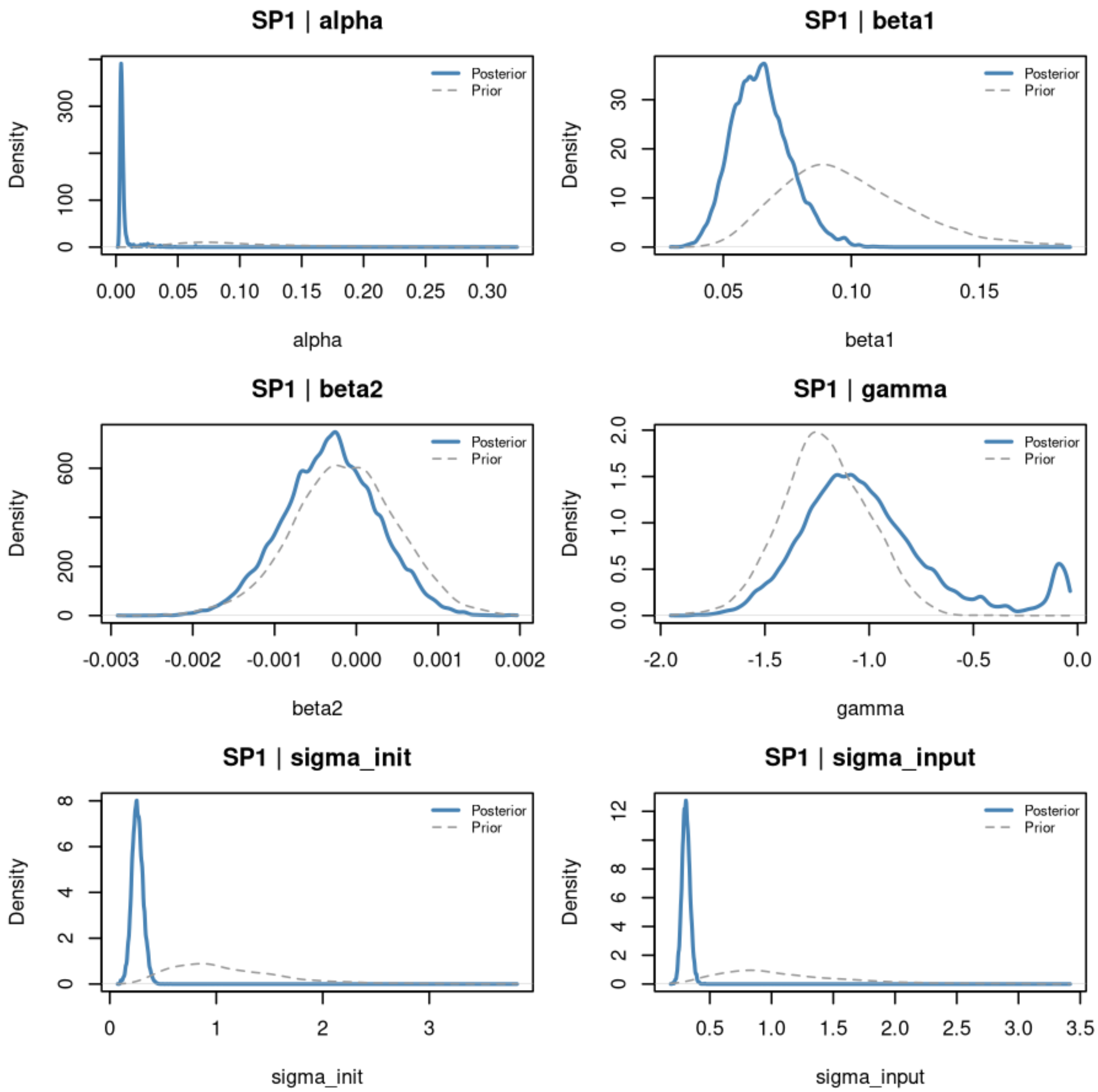


Figure 12: SP1: prior (dashed) vs posterior (solid) for all six free parameters.

15.2 TP2

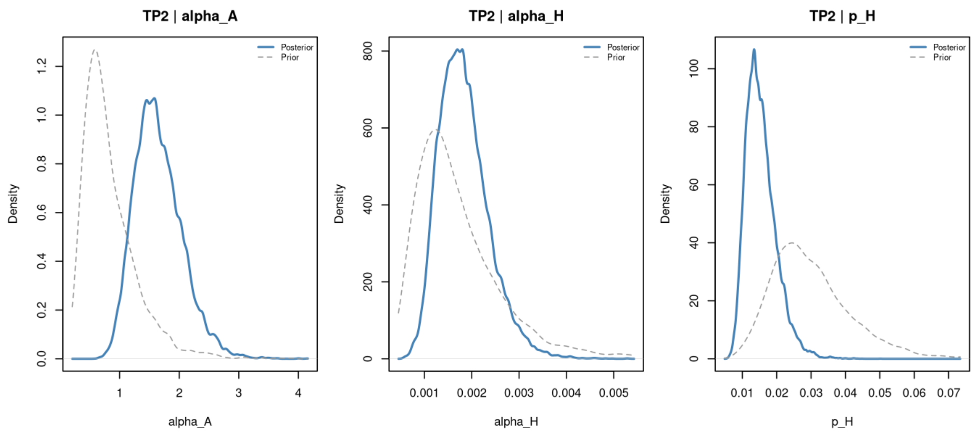


Figure 13: TP2 transfer/humification fractions: prior vs posterior.

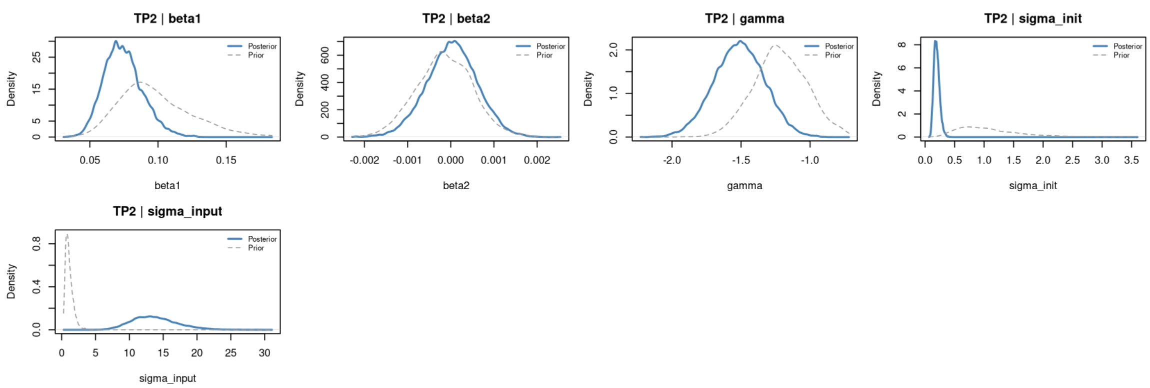


Figure 14: TP2 climate and auxiliary parameters: prior vs posterior.

15.3 TP3

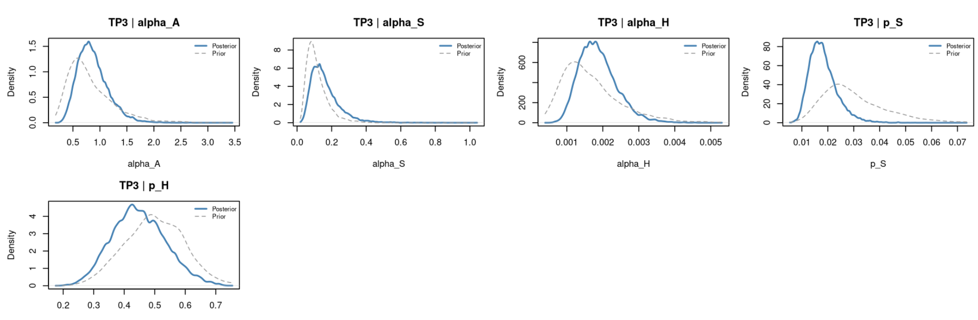


Figure 15: TP3 transfer fractions (p_S, p_H): prior vs posterior.

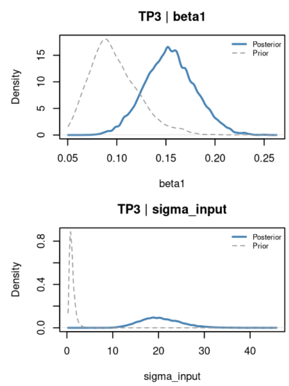


Figure 16: TP3 climate and auxiliary parameters: prior vs posterior.

15.4 Yasso07

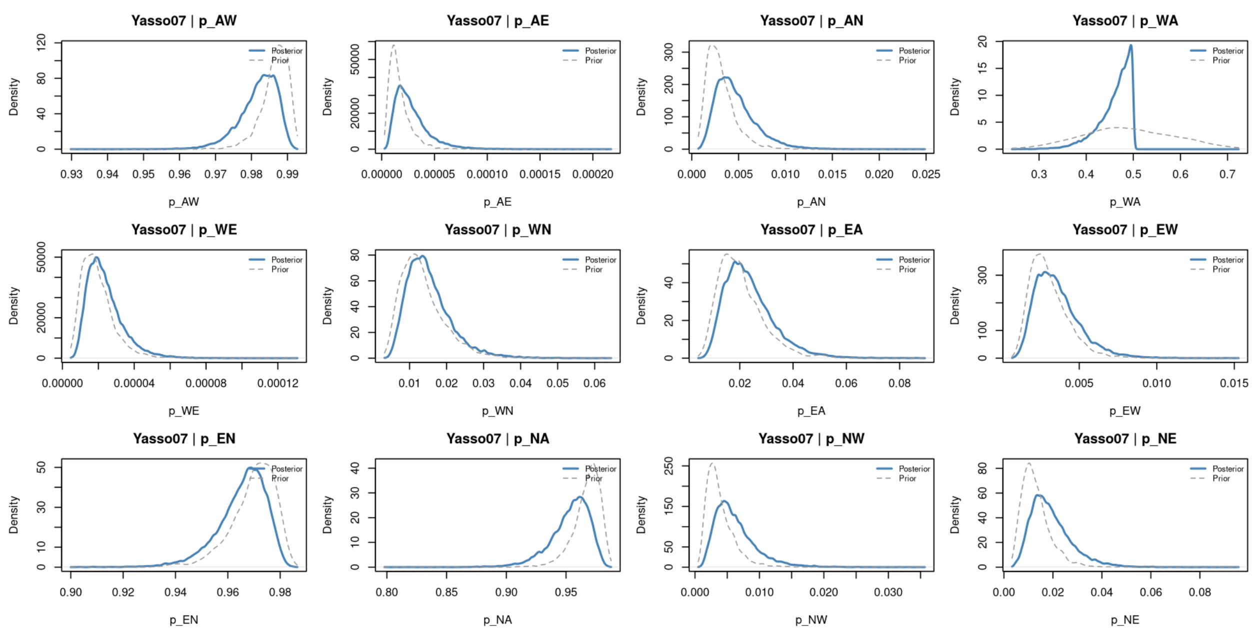


Figure 17: Yasso07 twelve transfer fractions: posterior ≈ prior throughout — the non-identifiability signature.

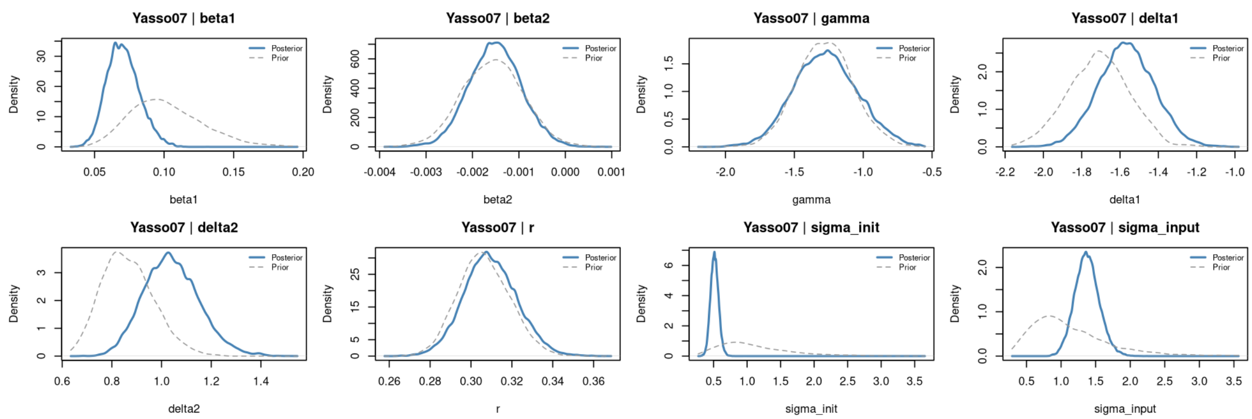


Figure 18: Yasso07 climate, size and auxiliary parameters: prior vs posterior.

15.5 Yasso15

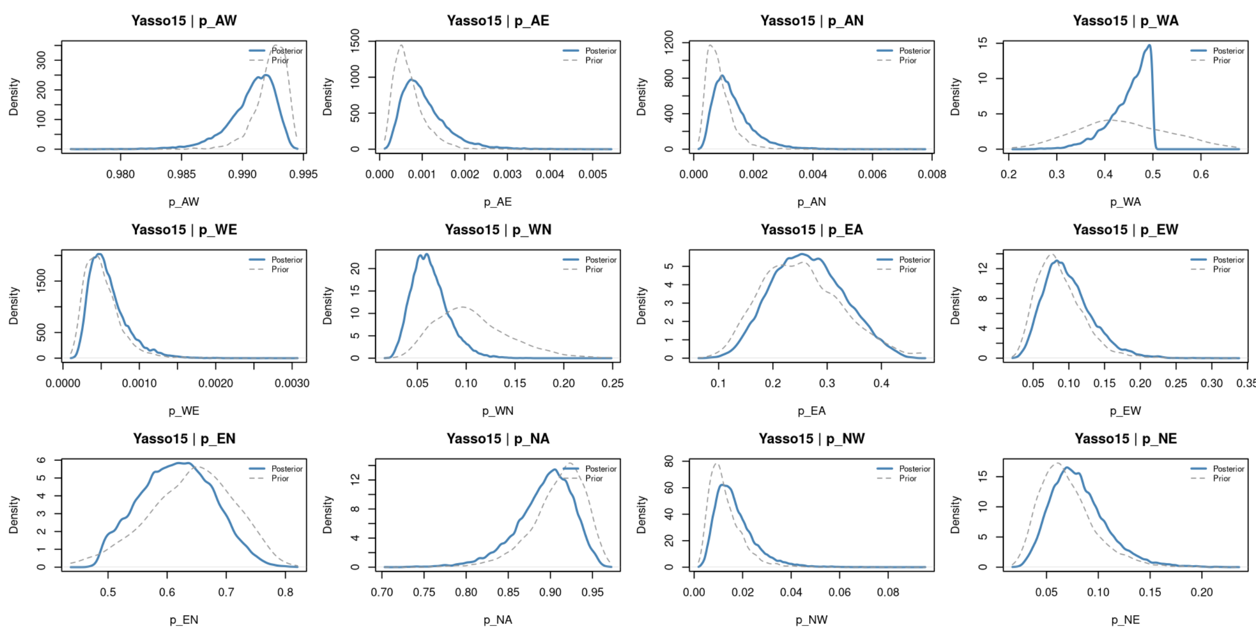


Figure 19: Yasso15 twelve transfer fractions: prior vs posterior.

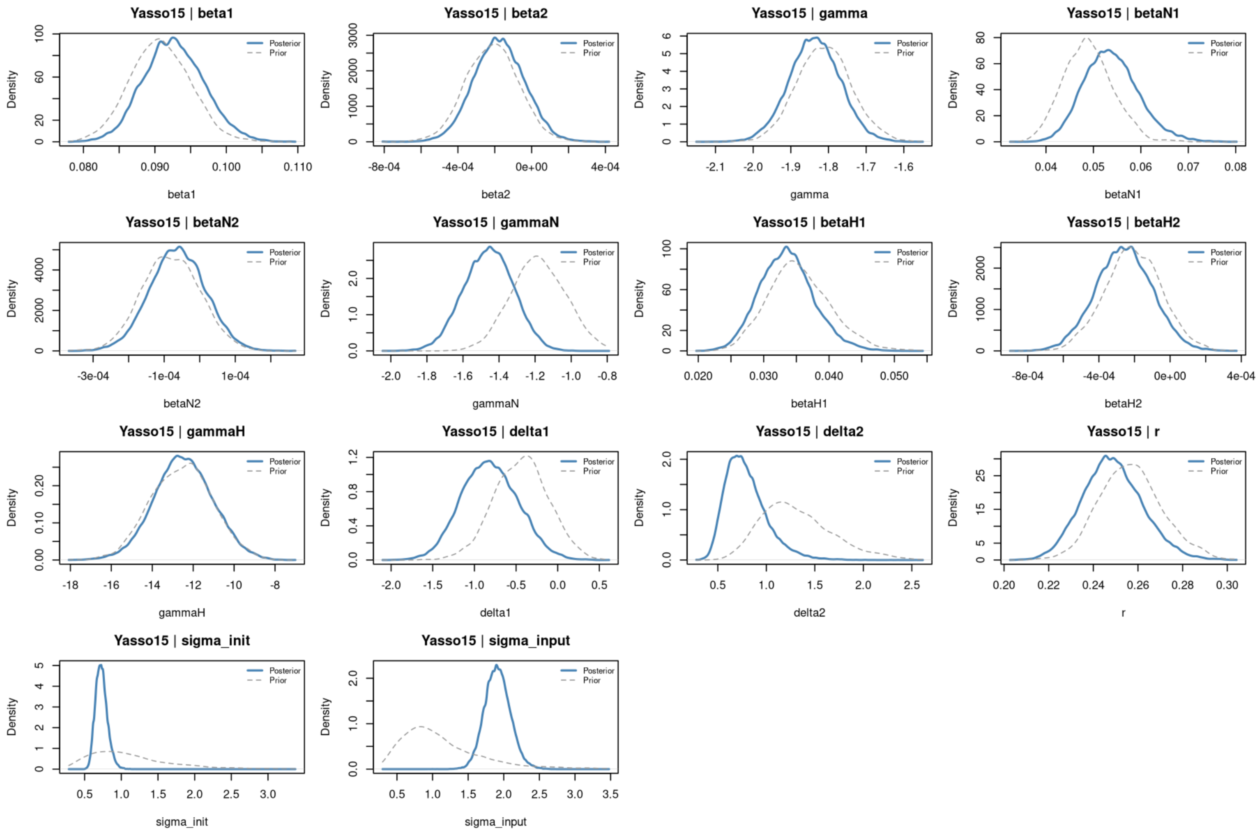


Figure 20: Yasso15 climate, size and auxiliary parameters: prior vs posterior.

15.6 Yasso20

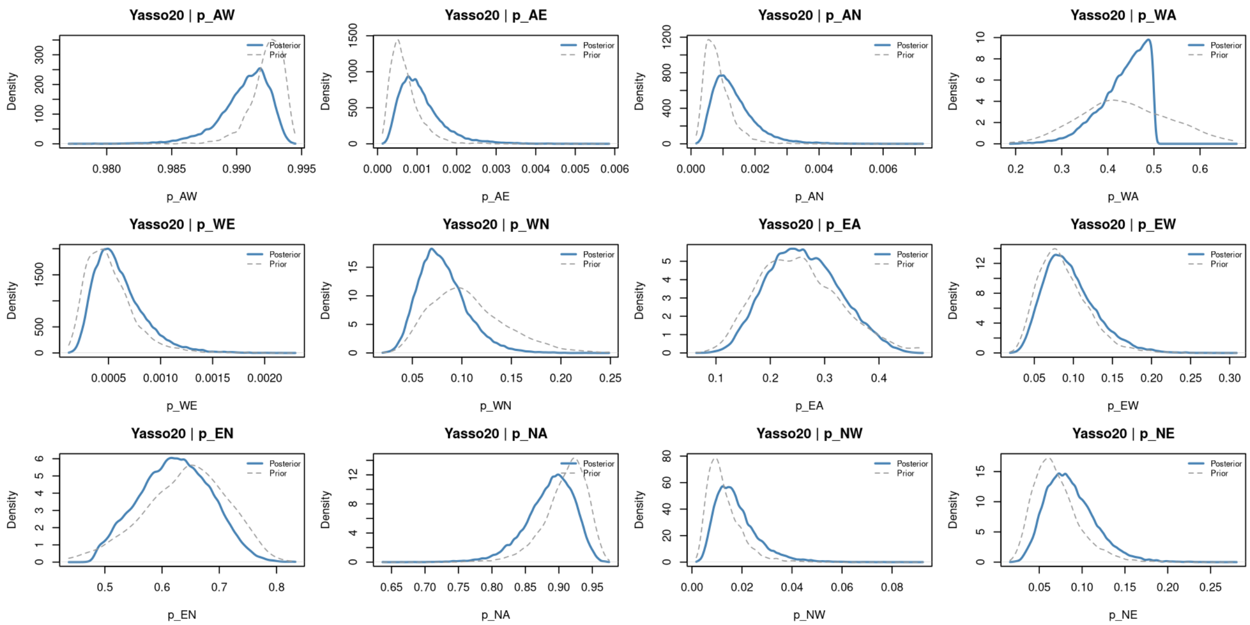


Figure 21: Yasso20 twelve transfer fractions: prior vs posterior. Despite all twelve being free, none is appreciably informed by the data.

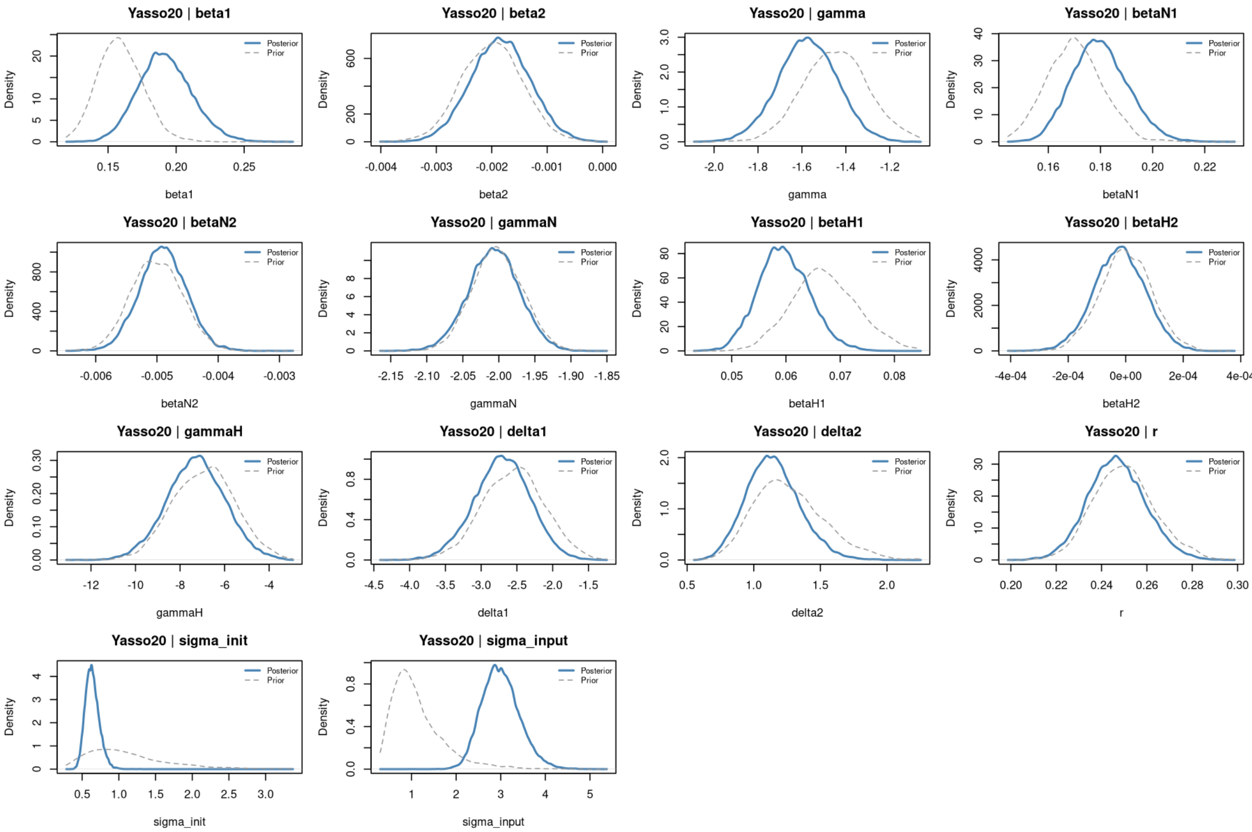


Figure 22: Yasso20 climate, size and auxiliary parameters: prior vs posterior.

---

16 Software

Component	Details
R	≥ 4.3
`BayesianTools`	Hartig et al. (2023); DEzs sampler
`coda`	Gelman–Rubin $\hat{R}$, ESS
`compiler`	`cmpfun()` byte-compilation of likelihood
`parallel`	`mclapply()` per-plot parallelism
`ranger`	Random-forest residual analysis
Fortran 90 cores	`R CMD SHLIB`; `.Fortran()` interface (Yasso family)
SP1, TP2, TP3	Pure R; no Fortran

Random seed: `set.seed(2025)` before chain initialisation, consistent across all models. Test plot subsampling (when `N_PLOTS_TEST` is set): `set.seed(42)`.

---

17 References

- Andrén, O. & Kätterer, T. (1997). ICBM: The introductory carbon balance model for exploration of soil carbon balances. *Ecological Applications* 7, 1226–1236.

- Hartig, F. et al. (2023). BayesianTools: General-purpose MCMC and SMC samplers and tools for Bayesian statistics. R package version 0.1.8.
- Tuomi, M. et al. (2008). Leaf litter decomposition – estimates of global variability based on Yasso07 model. *Ecological Modelling* 220, 3362–3371.
- Tuomi, M. et al. (2009). Heterotrophic soil respiration — comparison of different models on data from litterbag microenvironments. *Agriculture, Ecosystems and Environment* 132, 209–217.
- Tuomi, M. et al. (2011). Soil carbon model Yasso07 graphical user interface. *Environmental Modelling & Software* 26, 1358–1362.
- Tupek, B. et al. Litter input dataset for Finnish NFI plots. Zenodo. https://doi.org/10.5281/zenodo.19736499.
- Viskari, T. et al. (2022). Calibrating the soil organic carbon model Yasso20 with multiple Finnish datasets. *Geoscientific Model Development* 15, 1735–1752.